Project Report

On

"Quizara – Learn, Play, Achieve"

Submitted in Partial Fulfillment for Degree of

"Bachelor of Computer Application"

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VKS

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VIDHYADHAN CHARITABLE TRUST

SMT Z.S Patel College of Computer Application

Affiliated To

Veer Narmad South Gujarat University, Surat

Acknowledgement

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I would also like to extend my thanks to **SMT Z.S Patel College of Computer Application** for providing the necessary resources, facilities, and environment to carry out this work.

My heartfelt appreciation goes to my faculty members, friends, and classmates for their encouragement, suggestions, and assistance whenever needed.

Finally, I would like to thank my family for their constant motivation, patience, and moral support throughout this journey.

Place: _____ Signature: _____

Date: _____

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Chapter 1:
Introduction

1.1 Project Profile

Project Title	Quizara
Domain	Challenge Your Mind
Technology Stack	PHP/MySQL Stack (LAMP/WAMP/XAMPP)
Fronted	HTML5, CSS3, Vanilla JavaScript, Bootstrap 5 (v5.3.0)
Backend	Native PHP (v8.x recommended)
Database	MySQL (via PDO)
Tools	VS Code, Git, XAMPP

1.2 Overview of Project

Quizara serves as a comprehensive, dynamic web-based assessment platform specifically tailored for educational institutions, corporate training environments, and independent learners. In an era where continuous learning is paramount, Quizara bridges the gap between traditional test-taking and modern, interactive digital experiences. The platform is designed to offer a seamless quiz-taking experience that is both engaging for the student and deeply analytical for the administrator or educator. By leveraging established, reliable web technologies, it ensures that users enjoy fast load times, accessible interfaces, and a reliable connection regardless of their device.

The core philosophy behind Quizara is "Intuitive Assessment." This design principle influences every aspect of the user interface, from the clean typography and glassmorphism-inspired color palettes to the straightforward navigation and immediate feedback mechanisms. Unlike clunky, legacy Learning Management Systems (LMS) that often feel utilitarian and dated, Quizara prioritizes visual harmony and effective use of modern CSS trends. This approach not only elevates the perceived professionalism of the assessments but also reduces cognitive load for the test-taker, allowing them to focus entirely on demonstrating their knowledge.

From a technical perspective, Quizara is built upon the robust and time-tested PHP/MySQL stack, demonstrating a commitment to accessibility, wide server

compatibility, and rapid deployment. This architecture provides a straightforward development environment that enhances maintainability. The use of native PHP on the backend, communicating securely with a relational MySQL database via PHP Data Objects (PDO), ensures a stable, synchronous environment capable of handling user sessions, processing complex scoring algorithms, and generating dynamic content on the fly.

Security and data integrity are paramount in the Quizara ecosystem. The platform implements industry-standard security measures, including strictly managed PHP sessions and secure, time-limited tokens for password recovery, to ensure that user data is private. Sensitive data, such as user passwords, are hashed using bcrypt before storage, complying with best practices for data protection. Furthermore, the use of parameterized SQL queries via PDO ensures that the database is shielded from SQL injection vulnerabilities, maintaining the reliability of the assessment records.

Finally, Quizara is designed with functional expansion in mind. The architecture allows for the straightforward integration of new features, such as the recently added PHPMailer for automated notifications and the FPDF library for dynamic certificate generation. Whether it is adding new question types, integrating advanced reporting tools, or expanding the user base, the underlying system is engineered to handle increased complexity while maintaining its core focus on reliable, accessible online assessment.

1.3 Problem Statement

The primary challenge addressed by **Quizara** is the inefficiency and lack of engagement inherent in traditional, paper-based or rudimentary digital assessment methods. Conventional testing often involves significant manual effort for educators in terms of question distribution, time management, and the laborious process of grading and providing feedback. This manual overhead not only leads to delays in student performance analysis but also increases the likelihood of human error in scoring and record-keeping. Furthermore, many existing digital solutions are either too complex for quick deployment or lack the modern, intuitive user interfaces required to keep today's learners focused and motivated. The absence of immediate, visual feedback and tangible rewards often results in decreased student engagement and a less effective overall learning experience. Quizara seeks to resolve these issues by providing an automated, secure, and visually compelling platform that streamlines administrative workflows while maximizing the pedagogical value of every assessment.

Chapter 2:
Proposed System

2.1 Existing System & Limitations

Prior to the implementation of Quizara, the existing assessment framework largely relied on manual coordination and legacy digital tools that offered limited interactive capabilities. In the manual system, educators were responsible for the physical preparation and distribution of test materials, which was both time-consuming and environmentally unsustainable. Grading was a secondary, equally taxing phase, often requiring days to return results to students, thereby losing the critical window for immediate pedagogical intervention. While some basic digital tools were in use, they frequently suffered from rigid, non-responsive interfaces that performed poorly on mobile devices, and lacked robust security measures to prevent unauthorized access or data manipulation. Additionally, the absence of centralized data storage meant that tracking long-term student progress was a fragmented, difficult task. These limitations collectively hindered the scalability of assessment programs and failed to provide the dynamic, data-rich feedback loop necessary for modern educational environments.

2.2 Objectives

The primary objective of the Quizara platform is to modernize and streamline the process of knowledge assessment. In an educational landscape increasingly reliant on digital tools, having an intuitive and reliable testing platform is critical. Quizara aims to provide a visually engaging interface that encourages participation and reduces test anxiety. By moving away from paper-based or archaic digital testing systems, the platform seeks to create a dynamic environment that resonates with modern learners, thereby improving completion rates and assessment accuracy.

A key operational objective is to simplify the complex processes involved in quiz creation and result management. For administrators and educators, the platform is designed to be a powerful tool that simplifies daily tasks such as drafting multiple-choice questions, organizing content into specific categories, and grading attempts. By automating the scoring and organizing these workflows, Quizara aims to reduce manual grading errors, save educators immense amounts of time, and allow them to operate more efficiently.

Enhancing user engagement and providing immediate value are central to the platform's user-centric goals. Quizara strives to create a rewarding user journey, from the moment a student logs in to the instant they complete a quiz. This involves implementing clear progress indicators, immediate automated scoring on submission, and tangible rewards for success. By minimizing friction and offering immediate feedback, including downloadable PDF certificates for passing scores, the system encourages continued learning and improves overall user satisfaction.

Ensuring reliability and data integrity is a fundamental architectural objective. As an assessment tool, its software infrastructure must be able to securely handle concurrent test-takers without losing data or crashing. Quizara is engineered to be robust, utilizing the proven stability of Apache/Nginx web servers and relational MySQL databases. The goal is to provide a platform that institutions can trust to accurately record and maintain student performance histories over time.

Another significant objective is to ensure the security and privacy of student data. With the increasing requirements of educational privacy laws, building trust

with users is essential. Quizara aims to implement rigorous security protocols, including secure password hashing and token-based account recovery mechanisms. The objective is to safeguard sensitive student information, ensuring a secure environment for examination.

Lastly, the project aims to provide a data-driven foundation for educational decision-making. By capturing detailed information on quiz attempts, question accuracy, and overall scores, the system creates a valuable repository of performance data. The objective is to empower educators with insights that help them understand student comprehension, identify difficult topics, and tailor their teaching strategies for better educational outcomes.

2.3 Hardware & Software Platforms

Hardware Development Environment:

Minimum:	• **Processor:** Single-core Processor (1 GHz+) • **RAM:** 1GB (2GB recommended for local servers like XAMPP) • **Storage:** 100MB for application files, plus Database storage (HDD/SSD)
Recommended:	• **Processor**: Modern Multi-core Processor • **RAM:** 4GB minimum recommended • **Storage:** 512GB SSD for faster database I/O

Software Development Environment:

Operating System:	Windows 10/11, macOS 10.15+, or Linux.
Frontend	HTML5, CSS3, Vanilla JavaScript, Bootstrap 5 (v5.3.0)
Backend	Native PHP (v8.x recommended)
Database:	MySQL (via PDO)
Code Editor:	Visual Studio Code (Recommended).
Browser:	Chrome, Firefox, or Edge.
Version Control:	Git.

2.4 Scope of Project

The scope of Quizara encompasses a comprehensive Assessment Discovery module designed to facilitate an effortless learning experience. This includes a dynamic, paginated quiz catalog that allows students to browse through available tests with descriptions and metadata. The system supports category filtering and keyword search capabilities, enabling students to quickly find quizzes relevant to their current studies or interests, ensuring that the question bank is easily accessible and navigable.

User Management is another critical component within current scope. The system provides a secure and user-friendly registration and login process, featuring backend email validation and strong password hashing. Once authenticated, users have access to a personalized dashboard where they can manage their profiles, update security settings, and view their accumulated experience points (XP). The scope also includes a robust, secure "Forgot Password" flow utilizing 15-minute expiring tokens sent via email (PHPMailer integration).

Assessment Processing forms the core of the educational functionality. The project scope covers the entire lifecycle of a quiz attempt, starting from an interactive quiz interface that tracks time limits. This workflow collects user answers, processes them immediately upon submission against the database schema, and calculates a final score. Furthermore, the system tracks the complete attempt history for each user, providing them with a visual timeline of their past performances, pass/fail statuses, and the ability to review their detailed correct/incorrect answers.

On the administrative side, the scope includes a powerful Admin Dashboard. This restricted area provides educators with full control over the platform's content and operations. Admins can perform CRUD (Create, Read, Update, Delete) operations on Categories, Quizzes, and individual Questions (Multiple Choice, True/False). The dashboard also facilitates Student Management, allowing admins

to view registered users, monitor their activities, and manage platform-wide settings.

The security scope involves implementing fundamental measures to protect the platform and its users. This includes the implementation of strict PHP session management for stateful and secure access control, ensuring students cannot access admin routes. Input validation and sanitation on both the client (JS) and server (PHP) sides are included to prevent common web vulnerabilities. Specifically, all database interactions enforce Prepared Statements (PDO) to eradicate SQL injection risks.

Finally, the technical scope defines the boundaries of the platform's operation. Quizara is designed as a web-based responsive application accessible via standard web browsers on desktop, tablet, and mobile devices. It requires a standard web server equipped with PHP and MySQL. The current scope includes advanced reward mechanisms like on-the-fly PDF Certificate generation (via FPDF) for passing scores, and automated SMTP email notifications. It excludes, for this version, native mobile app development or live proctoring features.

2.5 Feasibility Study

The feasibility of Quizara was analyzed across three primary dimensions: technical, operational, and economic. Technically, the project leverages the well-established PHP/MySQL stack, which ensures broad compatibility with standard hosting environments and provides the necessary performance for handling concurrent users and complex database interactions. The integration of modern libraries like PHPMailer and FPDF is demonstrably achievable using native PHP, ensuring the system remains lightweight yet feature-rich. Operationally, the platform's intuitive design and automated grading workflows minimize the learning curve for administrators, ensuring high adoption rates within educational institutions. Economically, the use of open-source technologies (Apache, PHP, MySQL, Bootstrap) significantly reduces development and deployment costs, making Quizara a highly cost-effective solution for digital transformation in assessment. Overall, the project's reliance on mature technologies and clear user requirements makes it a highly feasible and sustainable endeavor.

Chapter 3:
System Design

3.1 System Architecture

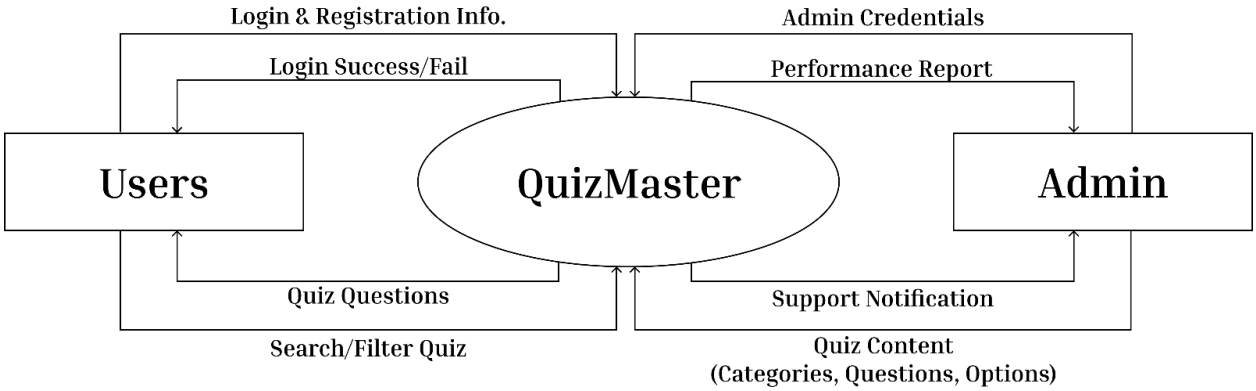
Quizara follows a classic three-tier architectural pattern consisting of a Presentation Tier, an Application Tier, and a Data Tier, ensuring clear separation of concerns and improved maintainability. The **Presentation Tier** is implemented using responsive HTML5, CSS3, and Bootstrap 5, providing the interactive user interface for both students and administrators across all devices. The **Application Tier** is powered by Native PHP, which handles the core business logic, including session management, authentication, and the complex grading algorithms. This layer acts as the bridge, processing user inputs from the frontend and executing corresponding queries against the database. Finally, the **Data Tier** utilizes a relational MySQL database to store persistent information such as user profiles, quiz questions, and attempt histories. This modular approach allows each tier to be expanded or updated independently, facilitating the future growth and scalability of the Quizara platform while maintaining a robust, secure foundation.

3.2 DFD (Data Flow Diagram)

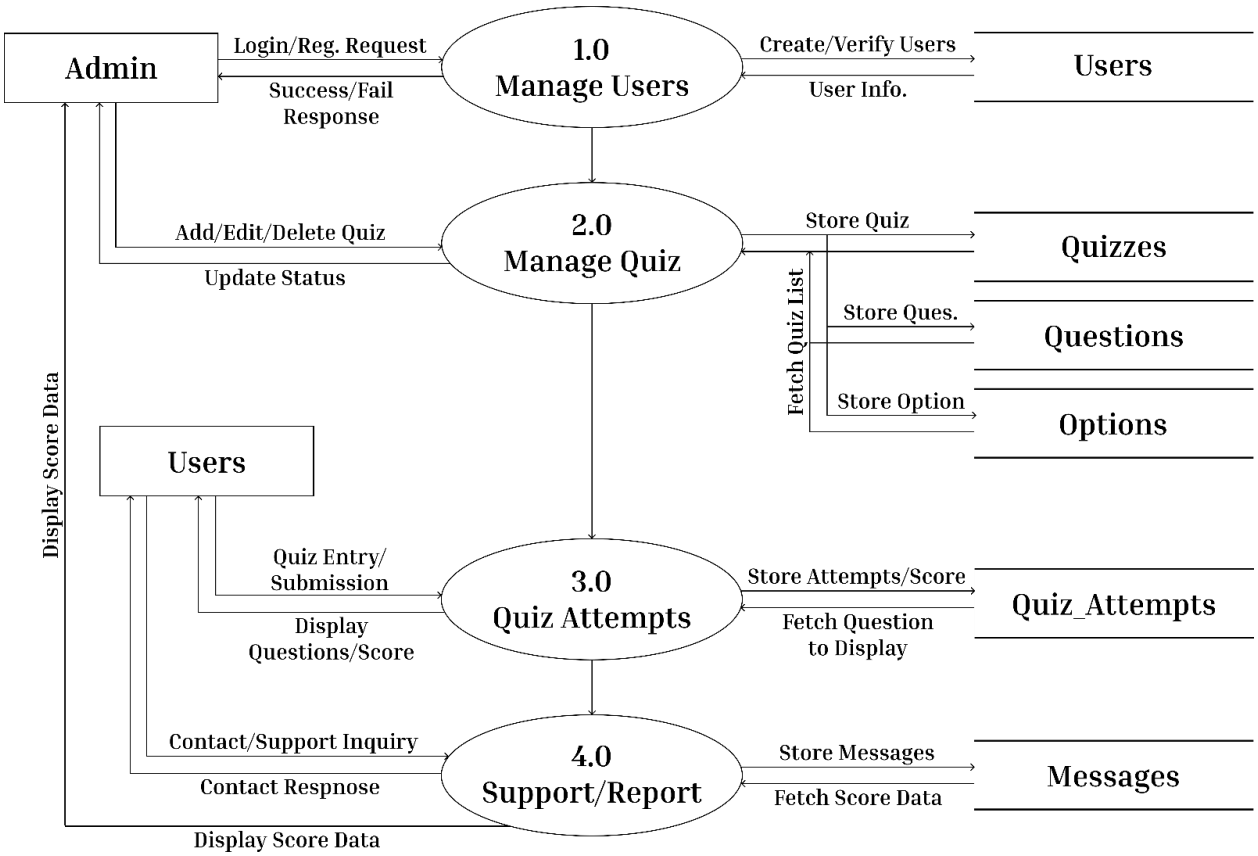
DFD Diagrams:

The Data Flow Diagram (DFD) serves as a visual representation of how information moves through the Quizara system. It maps out the flow of data from external entities, such as the Student and the Administrator, into the system's various processes and data stores.

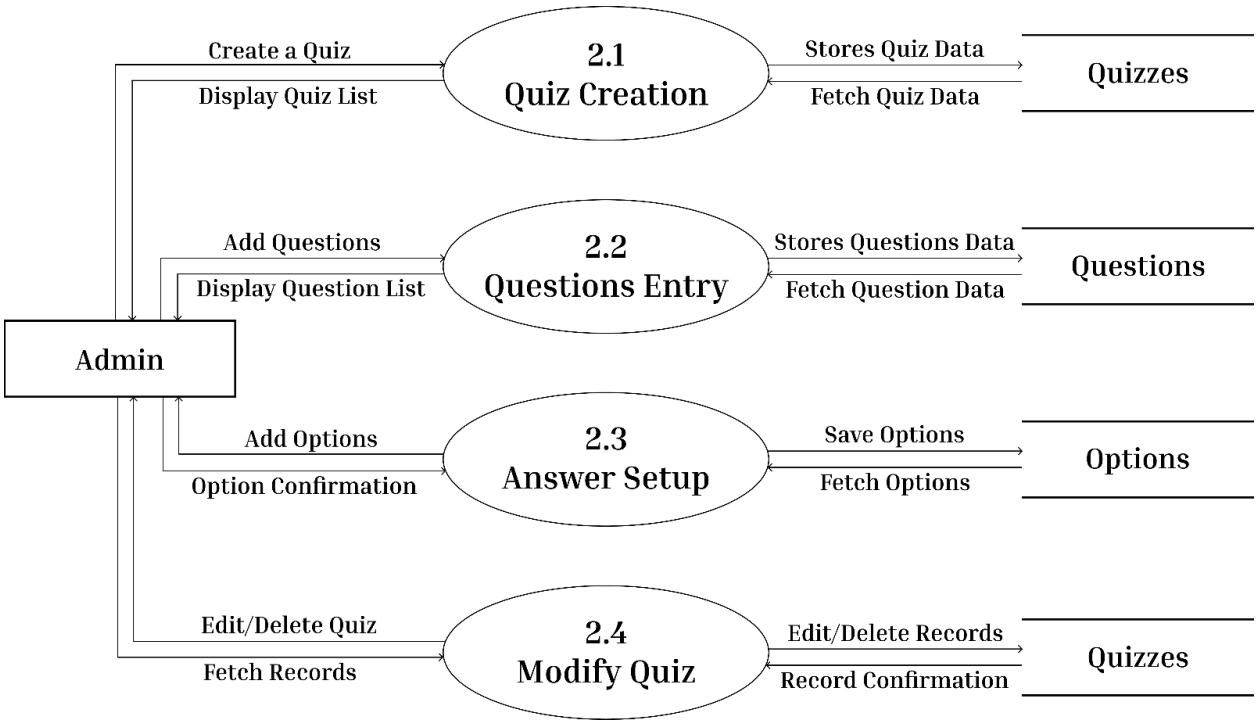
DFD Level 0:



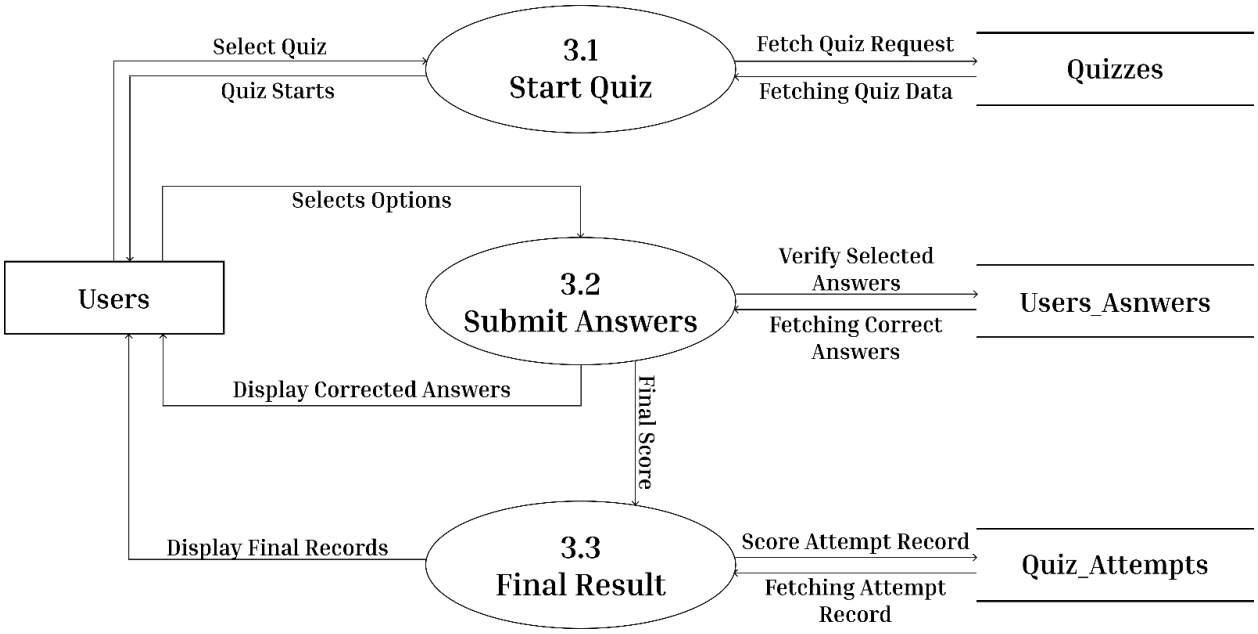
DFD Level 1:



DFD Level 2.1: (Manage Quiz)



DFD Level 3.1: (Quiz Attempts)

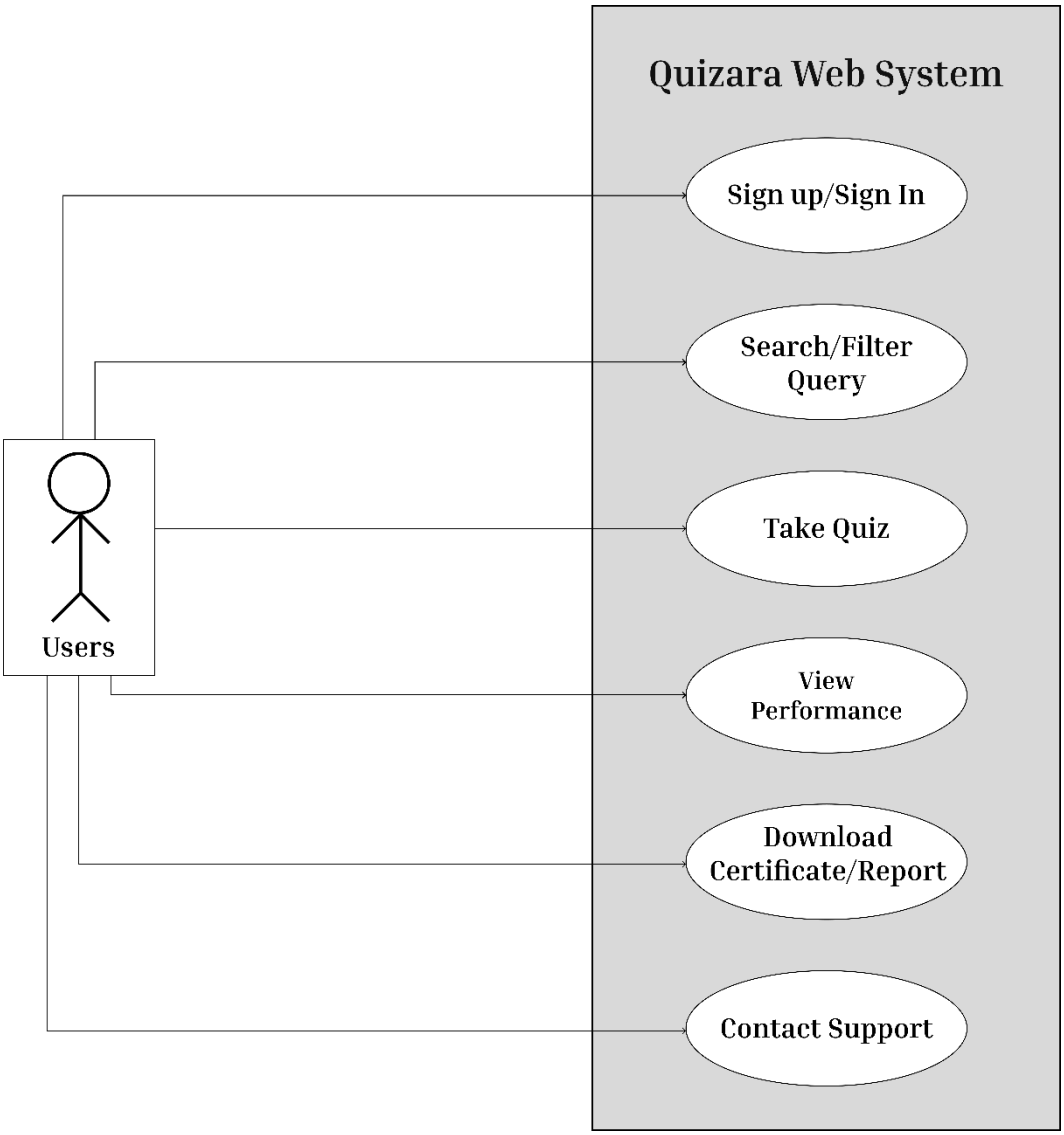


3.3 UML Diagram:

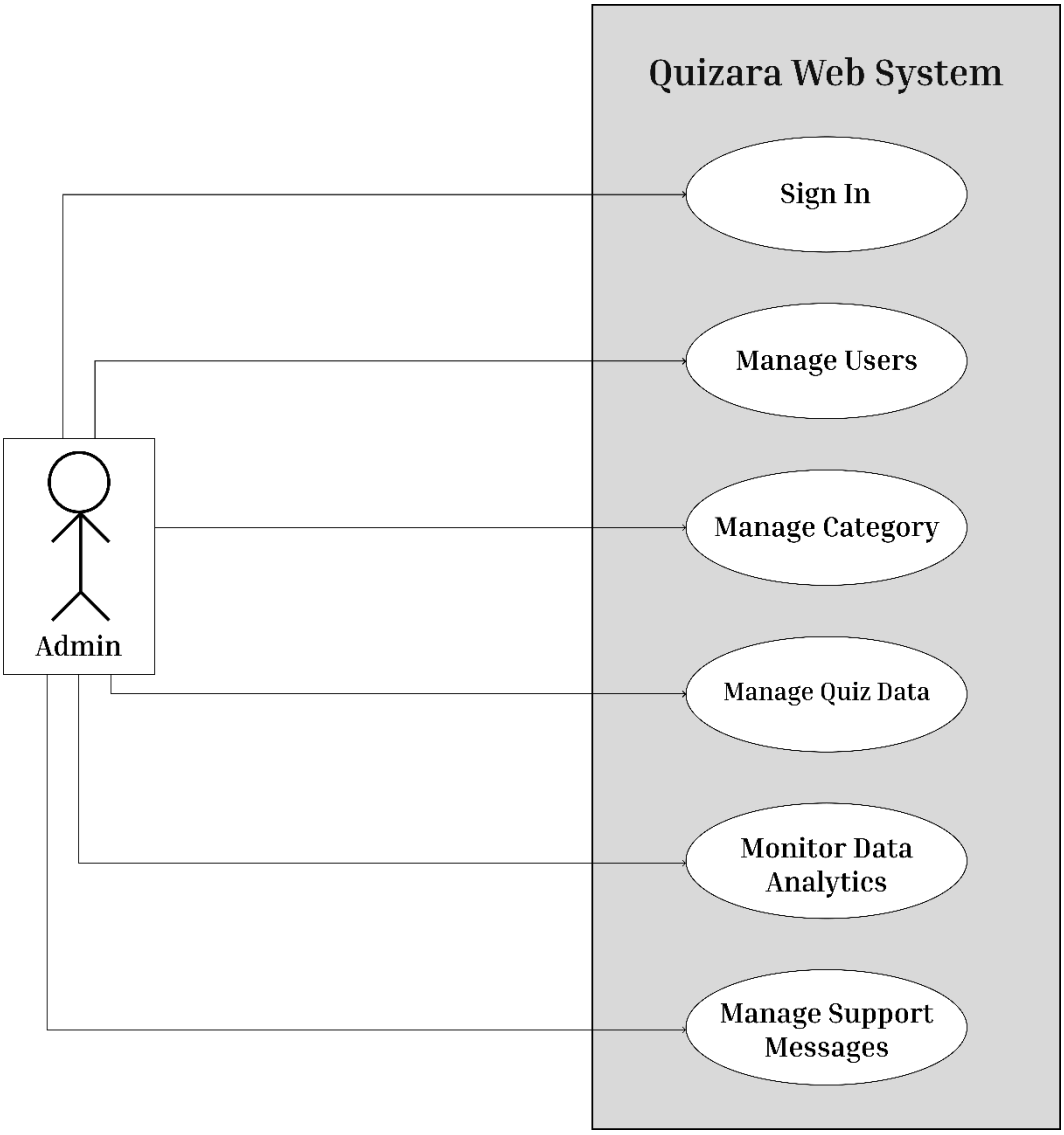
Use Case Diagrams:

In the context of our native PHP API, the UML diagram best translates to an Entity-Relationship (ER) model representing the relational database structure. This blueprint defines the tables, their attributes, and the strict foreign-key relationships between them, ensuring data integrity.

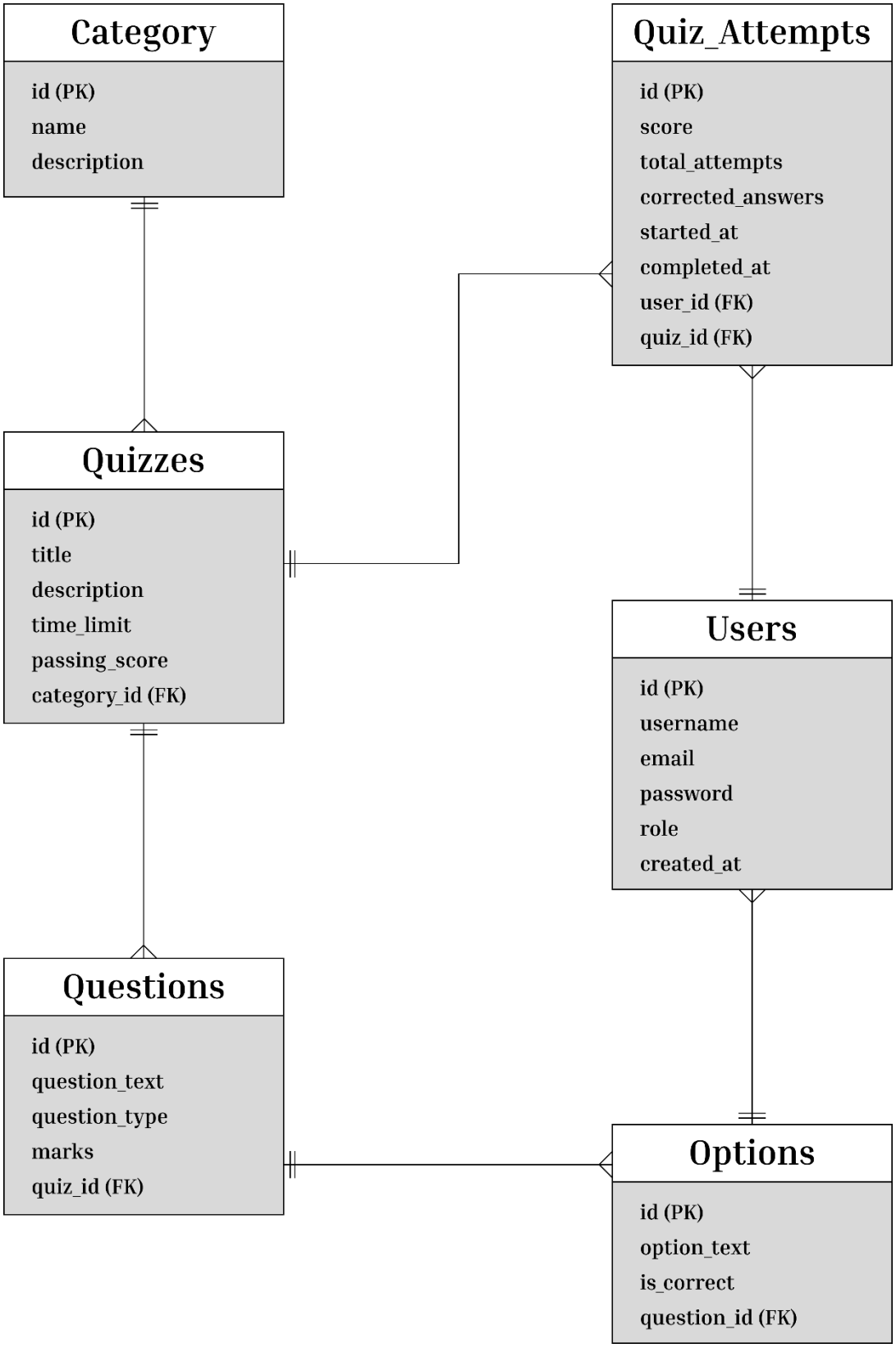
User's Use Case Diagram:



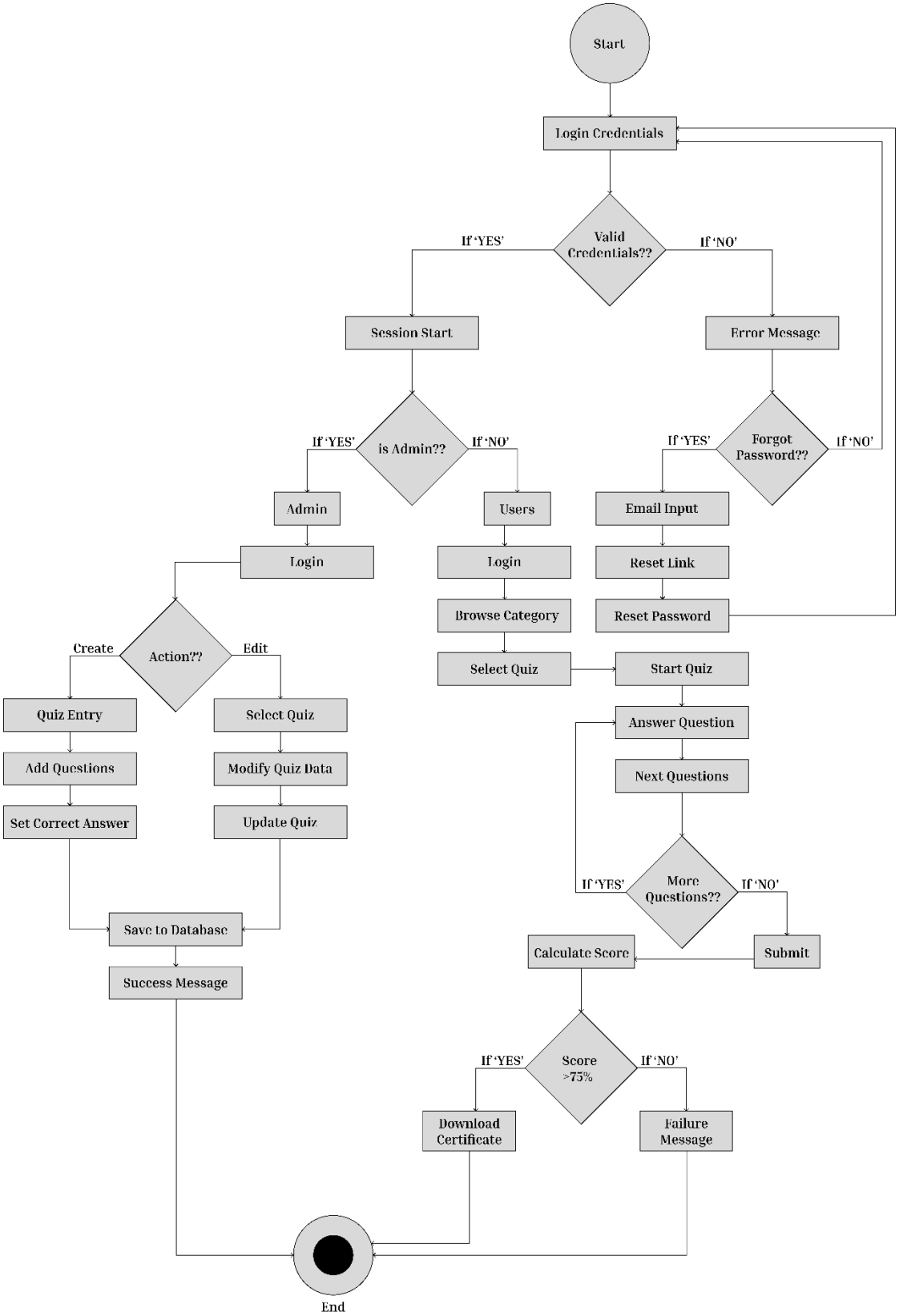
Use Case Diagram: (Admin)



3.3.1 ER Diagram



3.3.2 Activity Diagram



3.4 Data Dictionary

The Data Dictionary serves as a centralized repository of information about data, encompassing its meaning, relationships to other data, origin, usage, and format. In the context of the Quizara platform, it provides a structured and comprehensive breakdown of the core database collections implemented in MySQL. This documented vocabulary ensures that all developers and database administrators have a unified understanding of the data architecture, preventing inconsistencies in data definition and usage across different modules of the application.

1). Users Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each user.
username	VARCHAR(50)	Unique, Not Null	User-selected username used for login and identification.
email	VARCHAR(100)	Unique, Not Null	User's primary email address used for communication and password recovery.
password	VARCHAR(255)	Not Null	Hashed password stored using PHP password_hash() (bcrypt).
role	ENUM('admin','student')	Default: 'student'	Defines access level. Admin has full dashboard access; Student can take quizzes.
reset_token	VARCHAR(64)	Nullable	Secure random token used for password reset verification.
reset_token_expires_at	DATETIME	Nullable	Expiration timestamp (15 minutes validity) for password reset token.

2. Quizzes Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each quiz.
title	VARCHAR(255)	Not Null	Name of the quiz displayed to students.
description	TEXT	Nullable	Description of quiz subject or instructions.
category_id	INT	Foreign Key	References categories.id.
time_limit	INT	Default: 0	Time limit in minutes (0 = no time limit).
passing_score	INT	Default: 50	Minimum percentage required to pass the quiz.
created_by	INT	Foreign Key	References users.id (Admin who created the quiz).

3. Questions Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each question.
quiz_id	INT	Foreign Key	References quizzes.id (Cascade on delete).
question_text	TEXT	Not Null	The question displayed to the student.
question_type	ENUM('multiple_choice', 'true_false')	Default: 'multiple_choice'	Determines type of question rendering.
marks	INT	Default: 1	Marks awarded for correct answer.

4. Categories Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for category.
name	VARCHAR(100)	Not Null	Category name (e.g., Mathematics, Science).
description	TEXT	Nullable	Additional details about category.

5. Options Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each answer option.
question_id	INT	Foreign Key	References questions.id.
option_text	TEXT	Not Null	Text of the answer choice.
is_correct	TINYINT(1)	Default: 0	1 = Correct Answer, 0 = Incorrect Answer.

6. Quiz_Attempts Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each quiz attempt.
user_id	INT	Foreign Key	References users.id.
quiz_id	INT	Foreign Key	References quizzes.id.
score	INT	Default: 0	Final calculated score.
completed_at	TIMESTAMP	Nullable	Timestamp when quiz attempt was completed.

7. Users_Answers Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each answer submission.
attempt_id	INT	Foreign Key	References quiz_attempts.id.
question_id	INT	Foreign Key	References questions.id.
selected_option_id	INT	Foreign Key	References options.id.
is_correct	TINYINT(1)	Default: 0	Snapshot of correctness at submission time.

8. Settings Table:

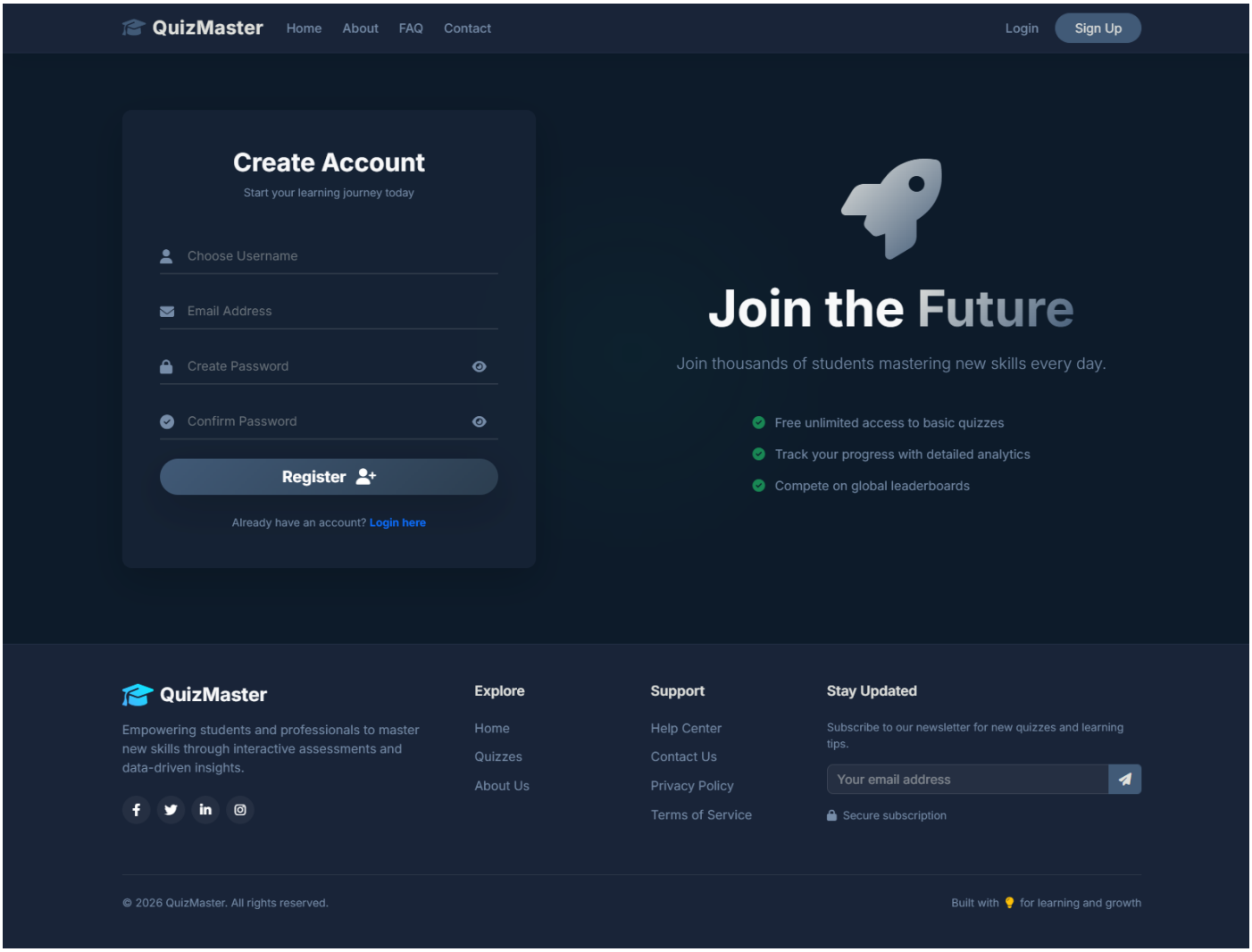
Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for configuration record.
setting_key	VARCHAR(50)	Unique, Not Null	Configuration key (e.g., site_name, maintenance_mode).
setting_value	TEXT	Nullable	Value assigned to the configuration key.
updated_at	TIMESTAMP	Auto Update	Automatically updates when record is modified.

9. Contact_Messages Table:

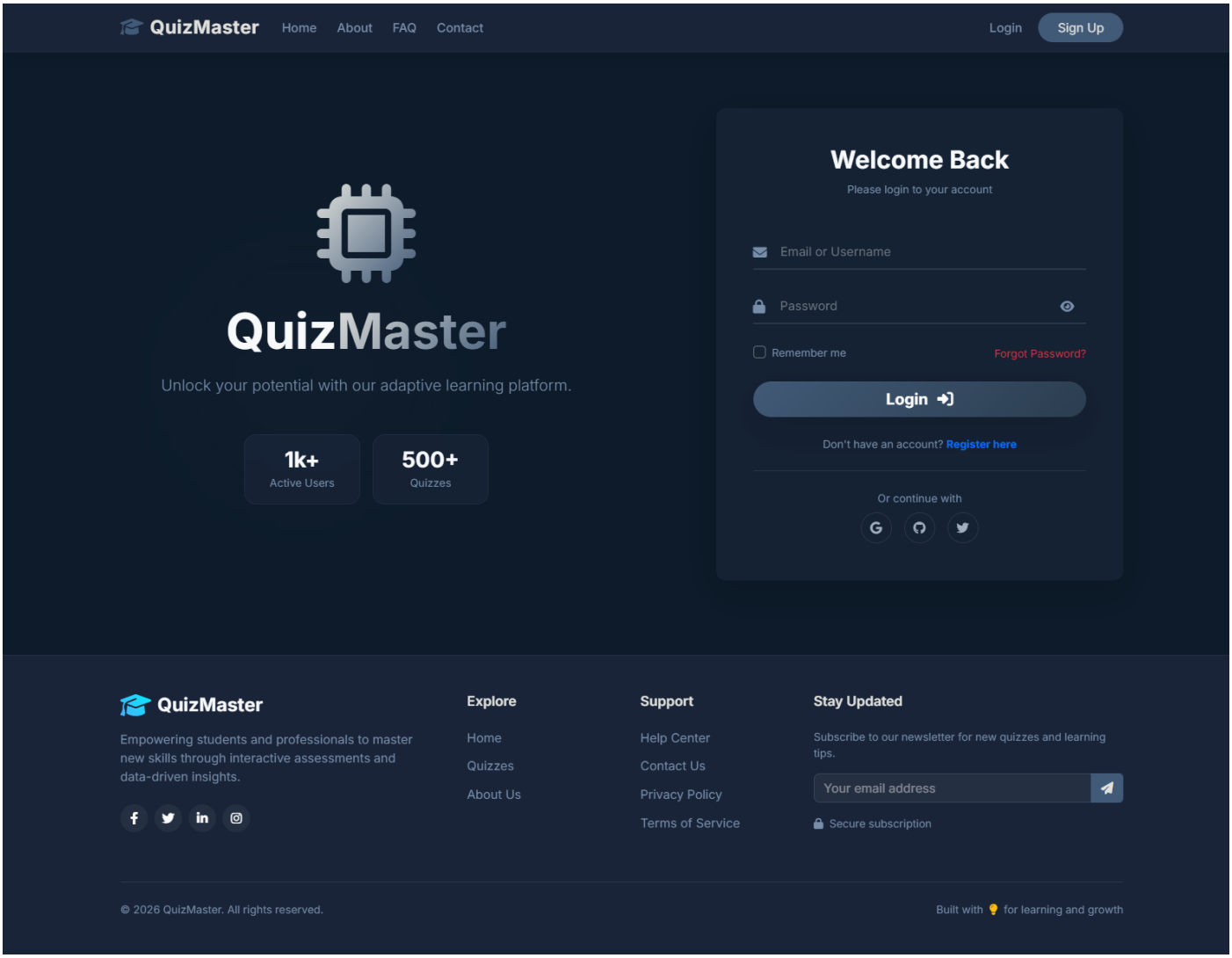
Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each contact message.
name	VARCHAR(100)	Not Null	Name of the person submitting the message.
email	VARCHAR(100)	Not Null	Email address for reply.
subject	VARCHAR(255)	Not Null	Subject/title of the message.
message	TEXT	Not Null	Detailed content of the message.
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Timestamp when message was submitted.

3.5 Interface (Screenshots)

Registration Page:



Login Page: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Unlock your potential with our adaptive learning platform.
1k+
Active Users
500+
Quizzes
Welcome Back
Please login to your account
Email or Username
Password
Remember me
Forgot Password?
Login
Don't have an account? Register here
Or continue with
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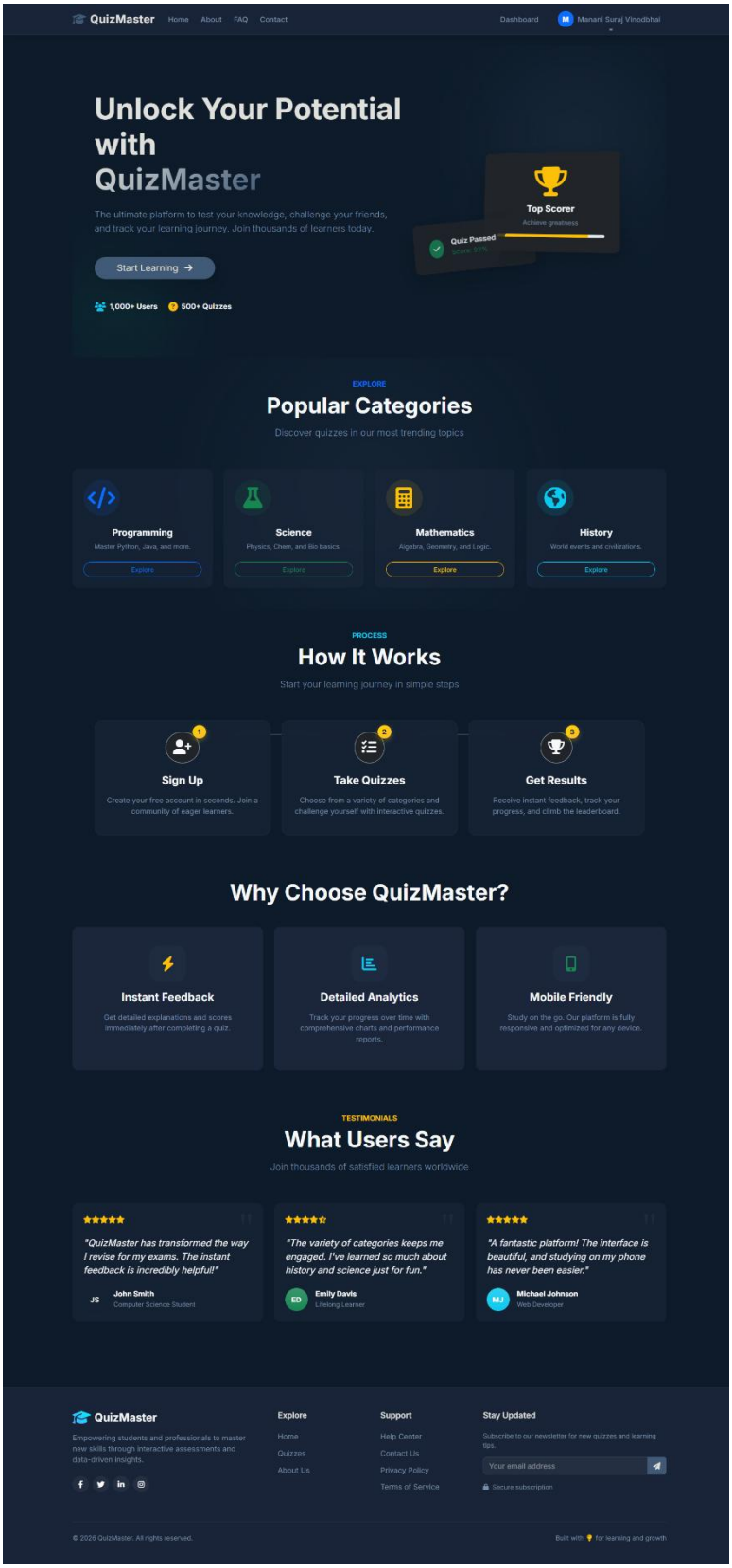
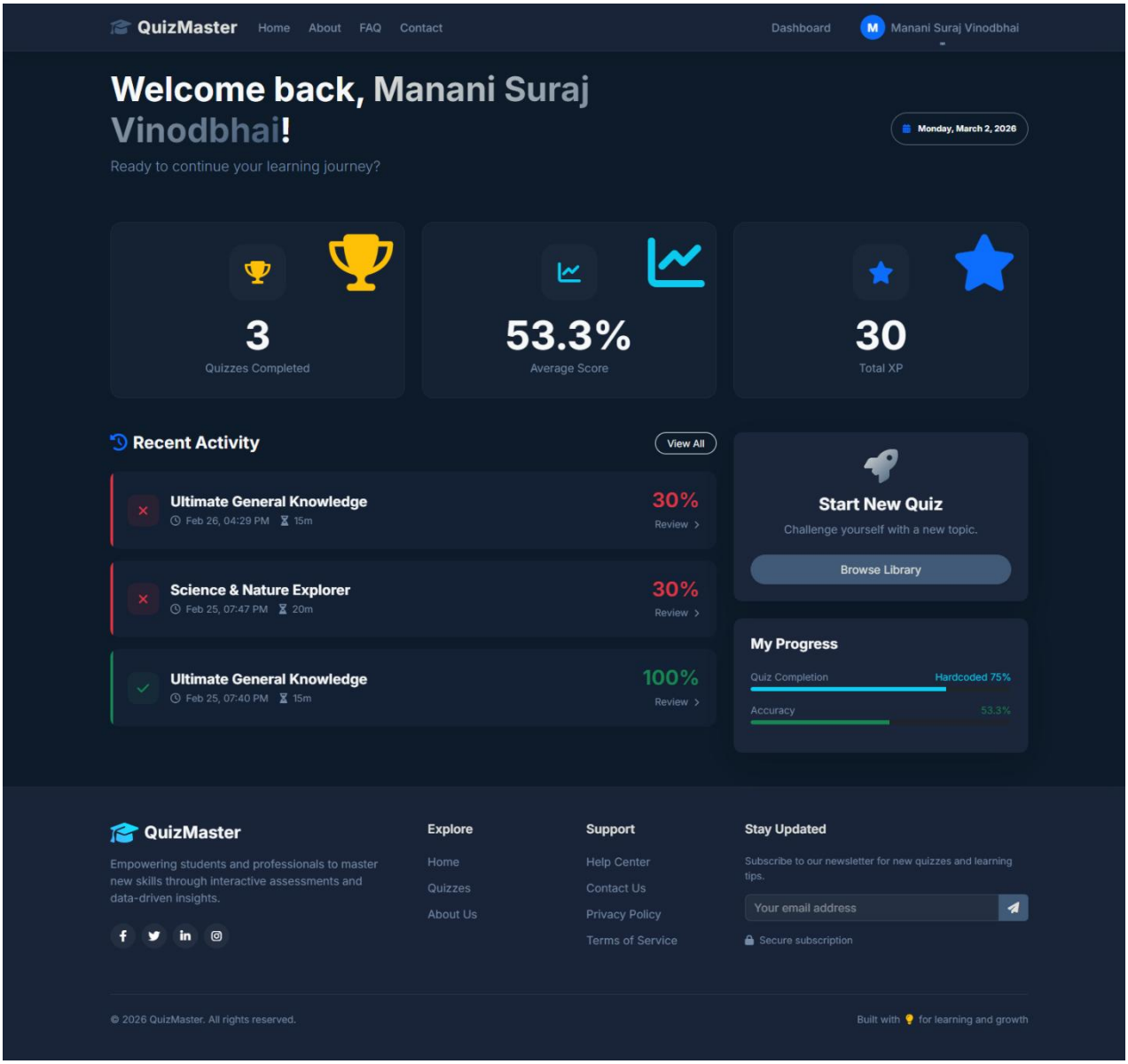
Home Page:



Dashboard Page:


QuizMaster
Home
About
FAQ
Contact
Dashboard
Manani Suraj Vinodbhai
Welcome back, Manani Suraj Vinodbhai!
Monday, March 2, 2026
Ready to continue your learning journey?
3
Quizzes Completed
53.3%
Average Score
30
Total XP
Recent Activity
View All
Ultimate General Knowledge
Feb 26, 04:29 PM
15m
30%
Review
Science & Nature Explorer
Feb 25, 07:47 PM
20m
30%
Review
Ultimate General Knowledge
Feb 25, 07:40 PM
15m
100%
Review
Start New Quiz
Challenge yourself with a new topic.
Browse Library
My Progress
Quiz Completion
Hardcoded 75%
Accuracy
53.3%
QuizMaster
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Quizzes
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Stay Updated
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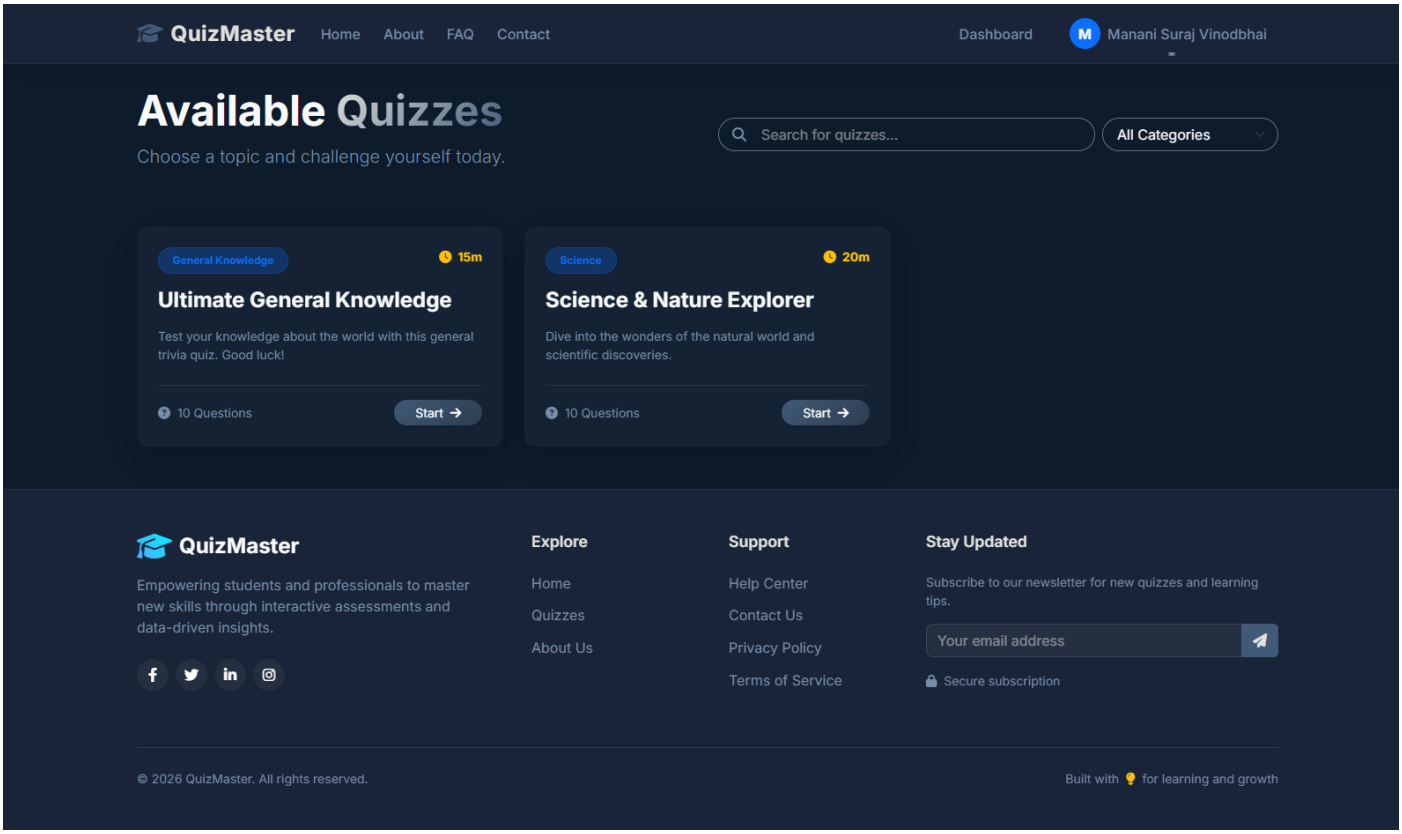
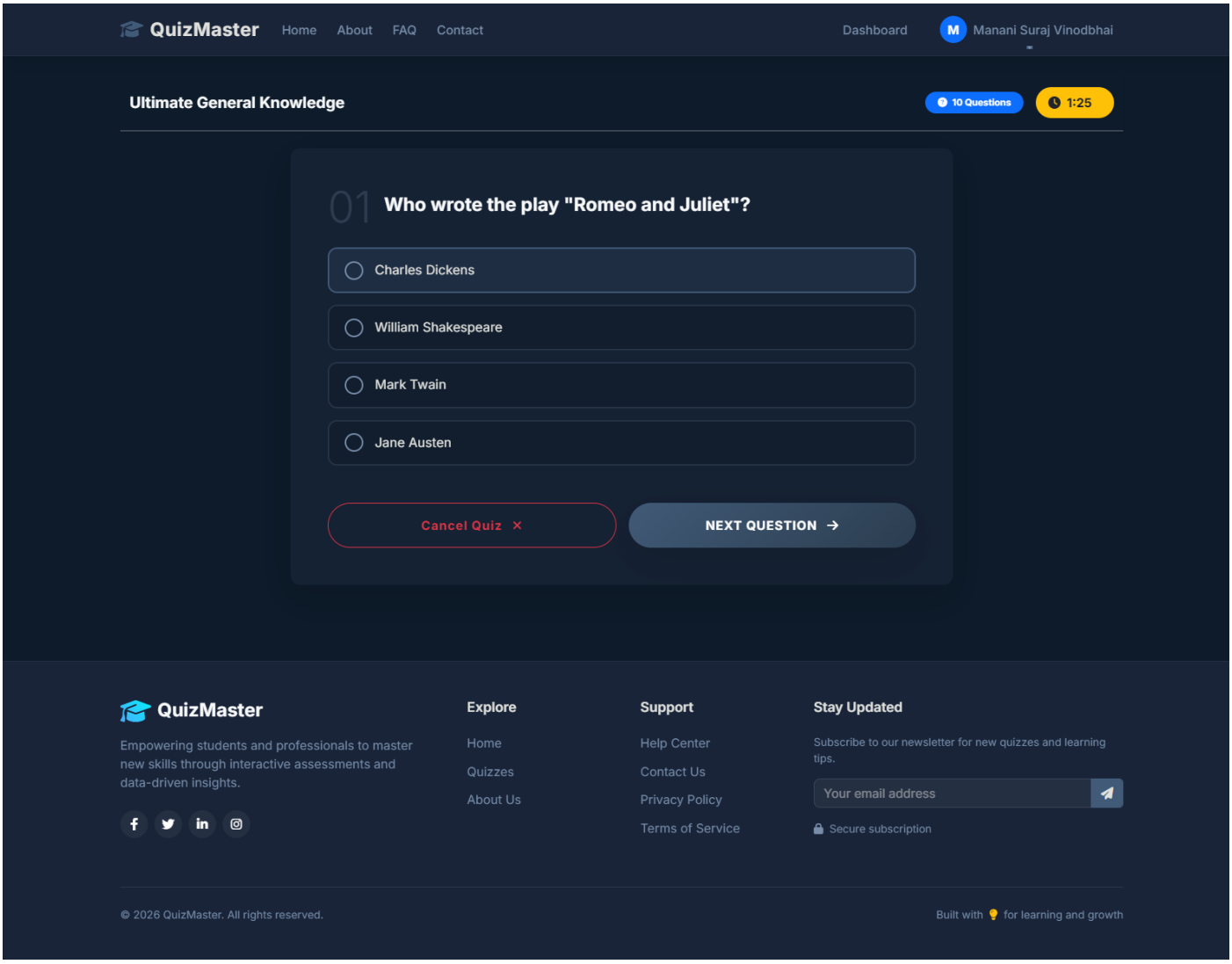
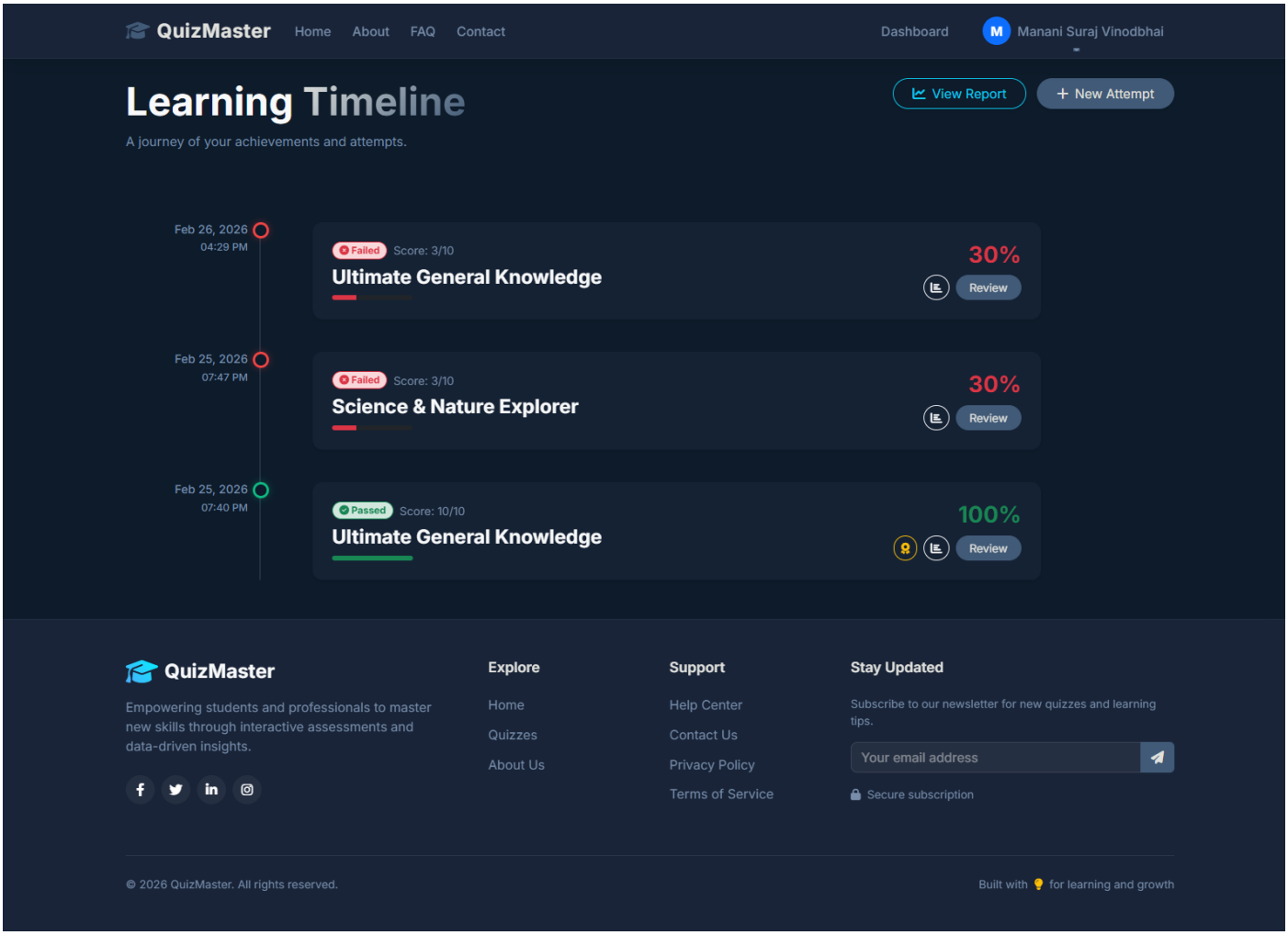
Quizzes Page:



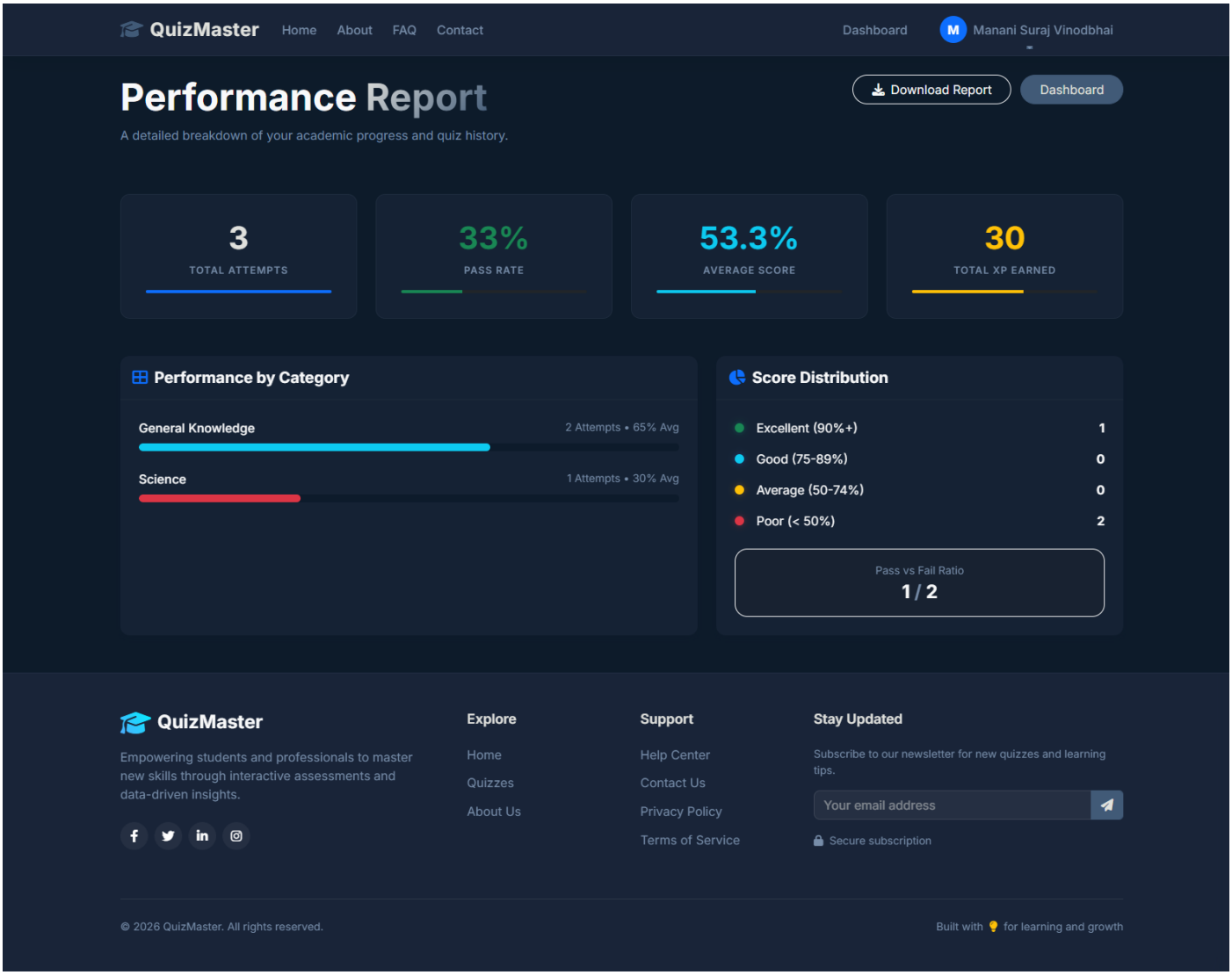
Take-Quiz Page:



History Page:



Report Page: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Contact
Dashboard
M
Manani Suraj Vinodbhai
Performance Report
A detailed breakdown of your academic progress and quiz history.
Download Report
Dashboard
3
TOTAL ATTEMPTS
33%
PASS RATE
53.3%
AVERAGE SCORE
30
TOTAL XP EARNED
Performance by Category
General Knowledge
2 Attempts • 65% Avg
Science
1 Attempts • 30% Avg
Score Distribution
Excellent (90%+)
1
Good (75-89%)
0
Average (50-74%)
0
Poor (< 50%)
2
Pass vs Fail Ratio
1 / 2
QuizMaster
Empowering students and professionals to master new skills through interactive assessments and data-driven insights.
Explore
Home
Quizzes
About Us
Support
Help Center
Contact Us
Privacy Policy
Terms of Service
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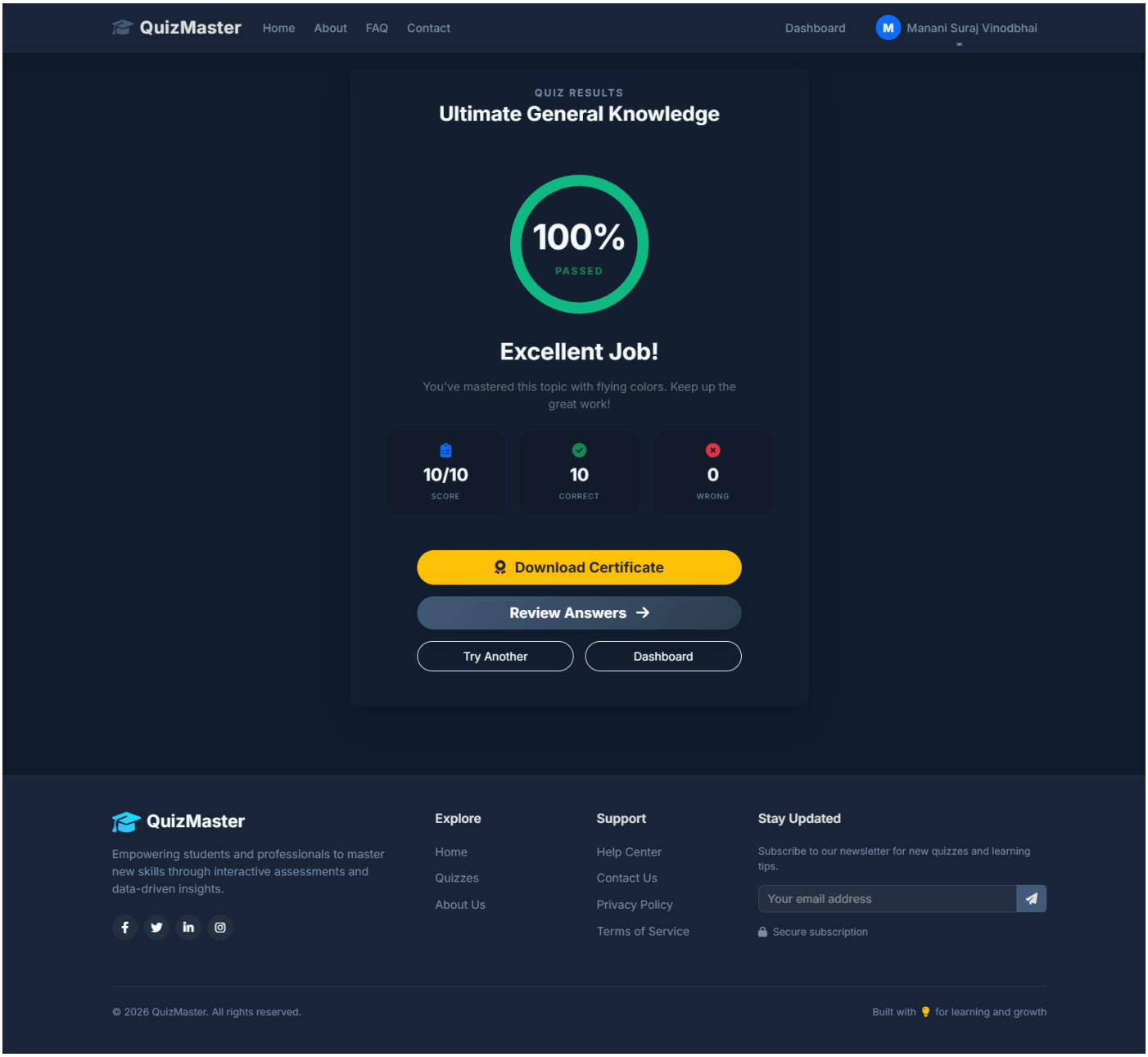
Result Page: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Dashboard
M
Manani Suraj Vinodbhai
QUIZ RESULTS
Ultimate General Knowledge
100%
PASSED
Excellent Job!
You've mastered this topic with flying colors. Keep up the great work!
10/10
SCORE
10
CORRECT
0
WRONG
Download Certificate
Review Answers →
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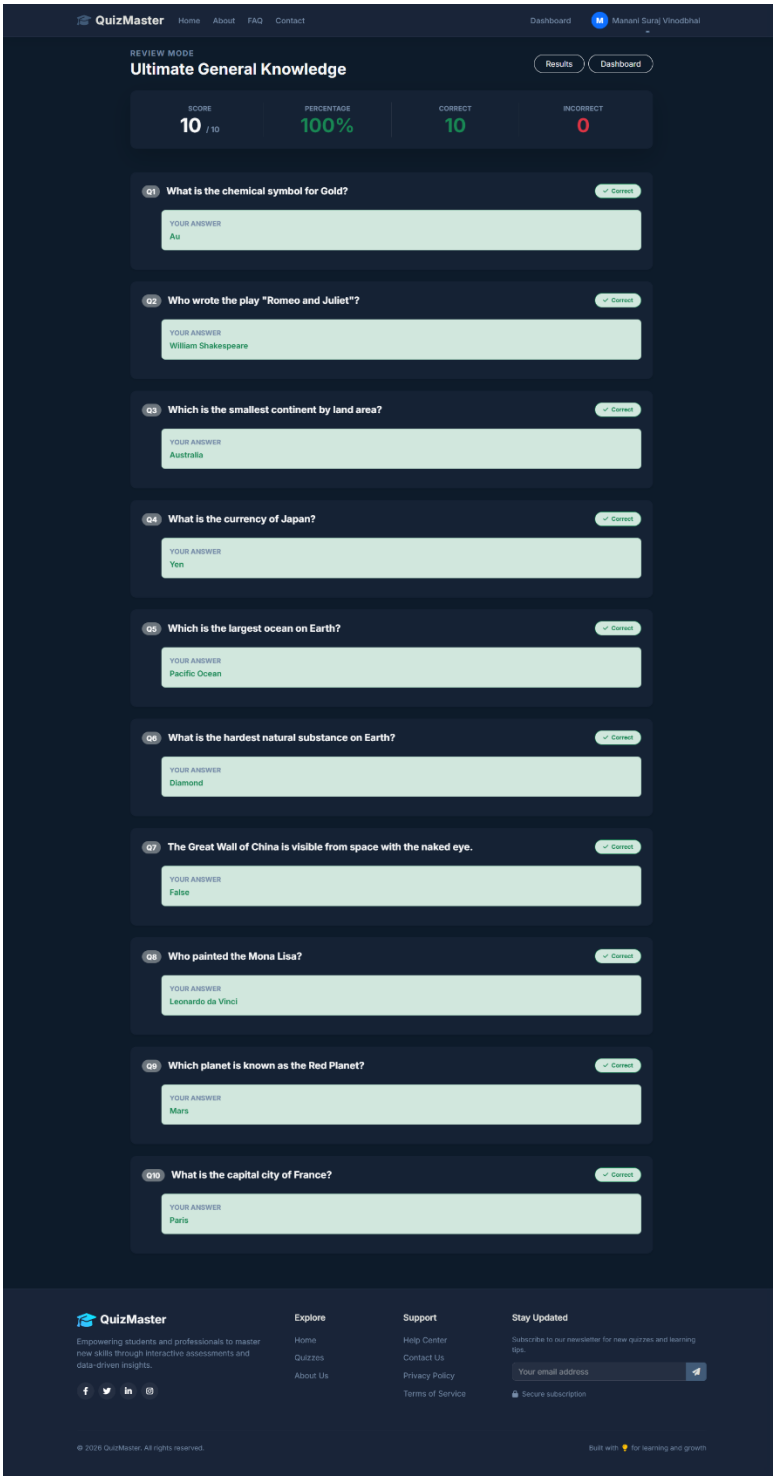
Review Page: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REVIEW MODE
Ultimate General Knowledge
Results
Dashboard
SCORE
10 / 10
PERCENTAGE
100%
CORRECT
10
INCORRECT
0
Q1 What is the chemical symbol for Gold?
Correct
YOUR ANSWER
Au
Q2 Who wrote the play "Romeo and Juliet"?
Correct
YOUR ANSWER
William Shakespeare
Q3 Which is the smallest continent by land area?
Correct
YOUR ANSWER
Australia
Q4 What is the currency of Japan?
Correct
YOUR ANSWER
Yen
Q5 Which is the largest ocean on Earth?
Correct
YOUR ANSWER
Pacific Ocean
Q6 What is the hardest natural substance on Earth?
Correct
YOUR ANSWER
Diamond
Q7 The Great Wall of China is visible from space with the naked eye.
Correct
YOUR ANSWER
False
Q8 Who painted the Mona Lisa?
Correct
YOUR ANSWER
Leonardo da Vinci
Q9 Which planet is known as the Red Planet?
Correct
YOUR ANSWER
Mars
Q10 What is the capital city of France?
Correct
YOUR ANSWER
Paris
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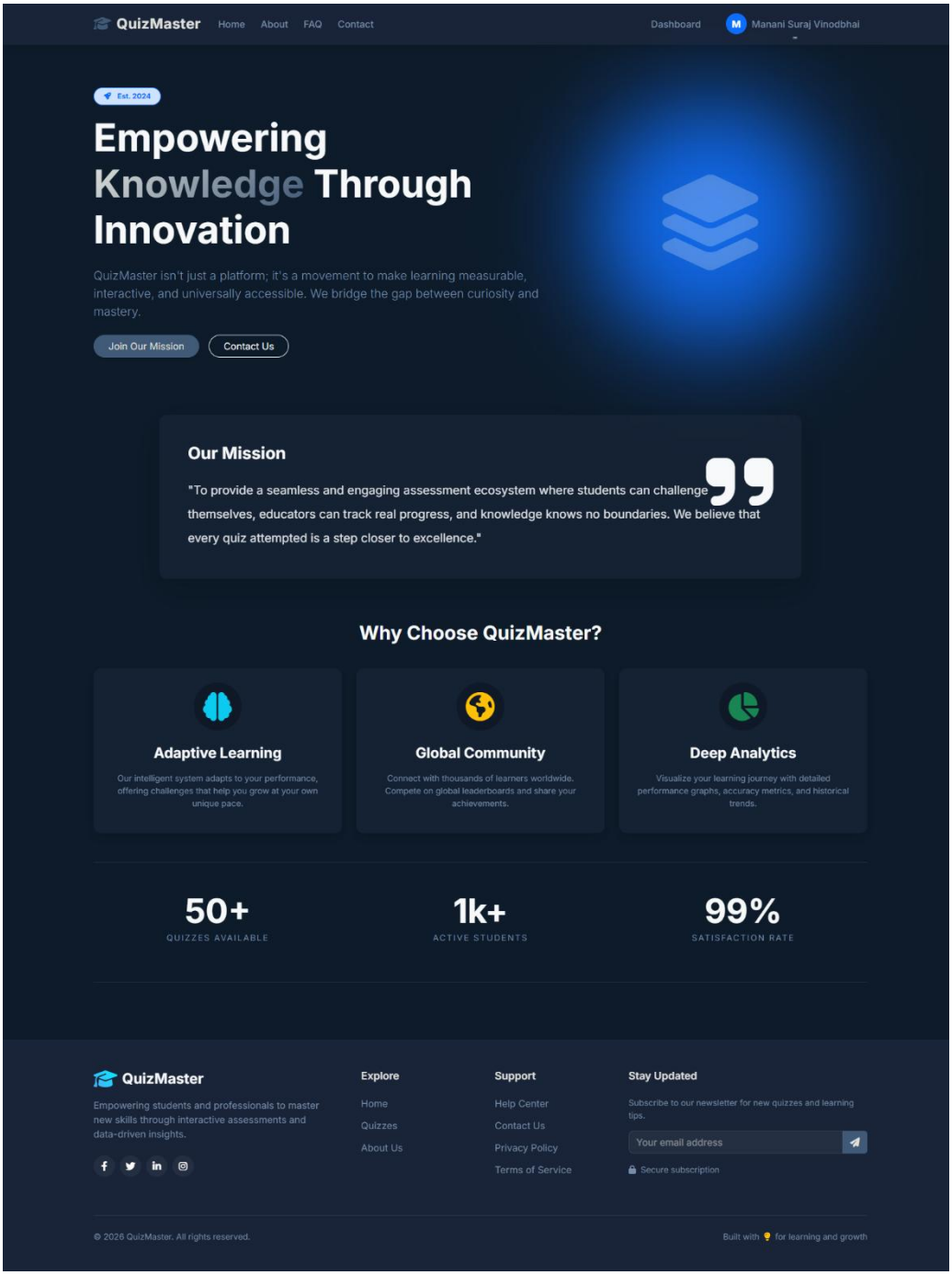
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Est. 2024
Empowering Knowledge Through Innovation
QuizMaster isn't just a platform; it's a movement to make learning measurable, interactive, and universally accessible. We bridge the gap between curiosity and mastery.
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Our Mission
"To provide a seamless and engaging assessment ecosystem where students can challenge themselves, educators can track real progress, and knowledge knows no boundaries. We believe that every quiz attempted is a step closer to excellence."
Why Choose QuizMaster?
Adaptive Learning
Our intelligent system adapts to your performance, offering challenges that help you grow at your own unique pace.
Global Community
Connect with thousands of learners worldwide. Compete on global leaderboards and share your achievements.
Deep Analytics
Visualize your learning journey with detailed performance graphs, accuracy metrics, and historical trends.
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99%
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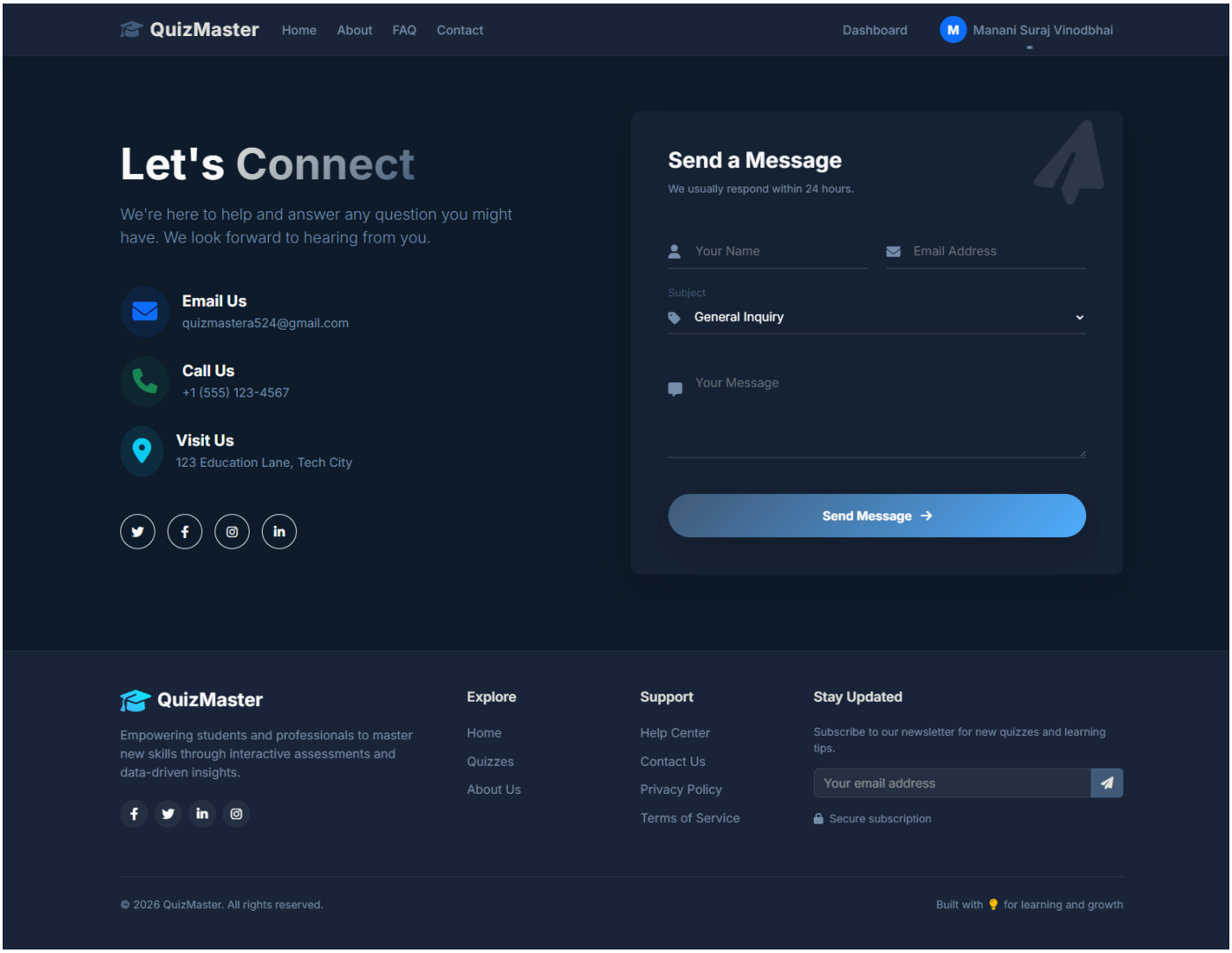
Contact Page:



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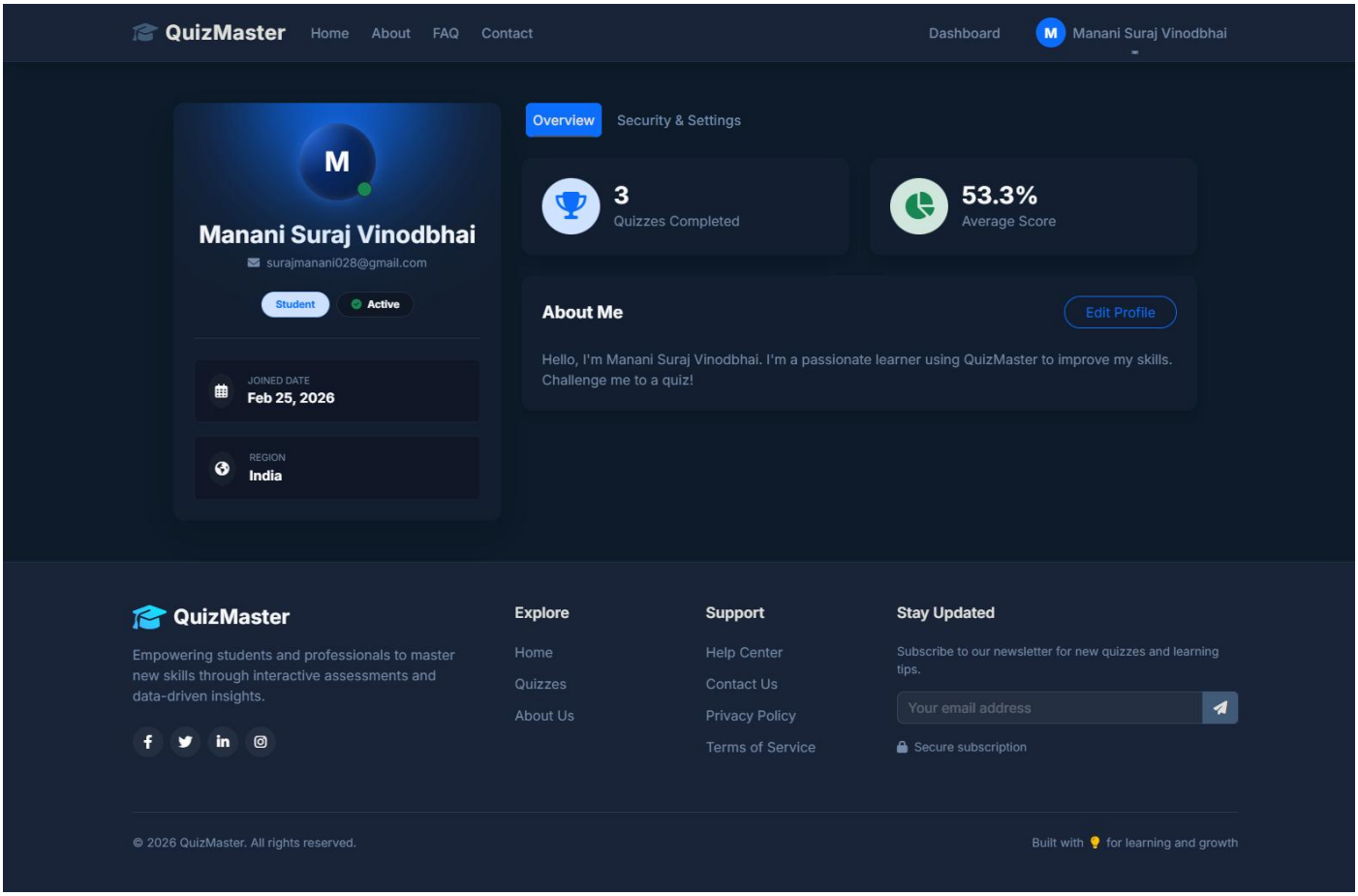

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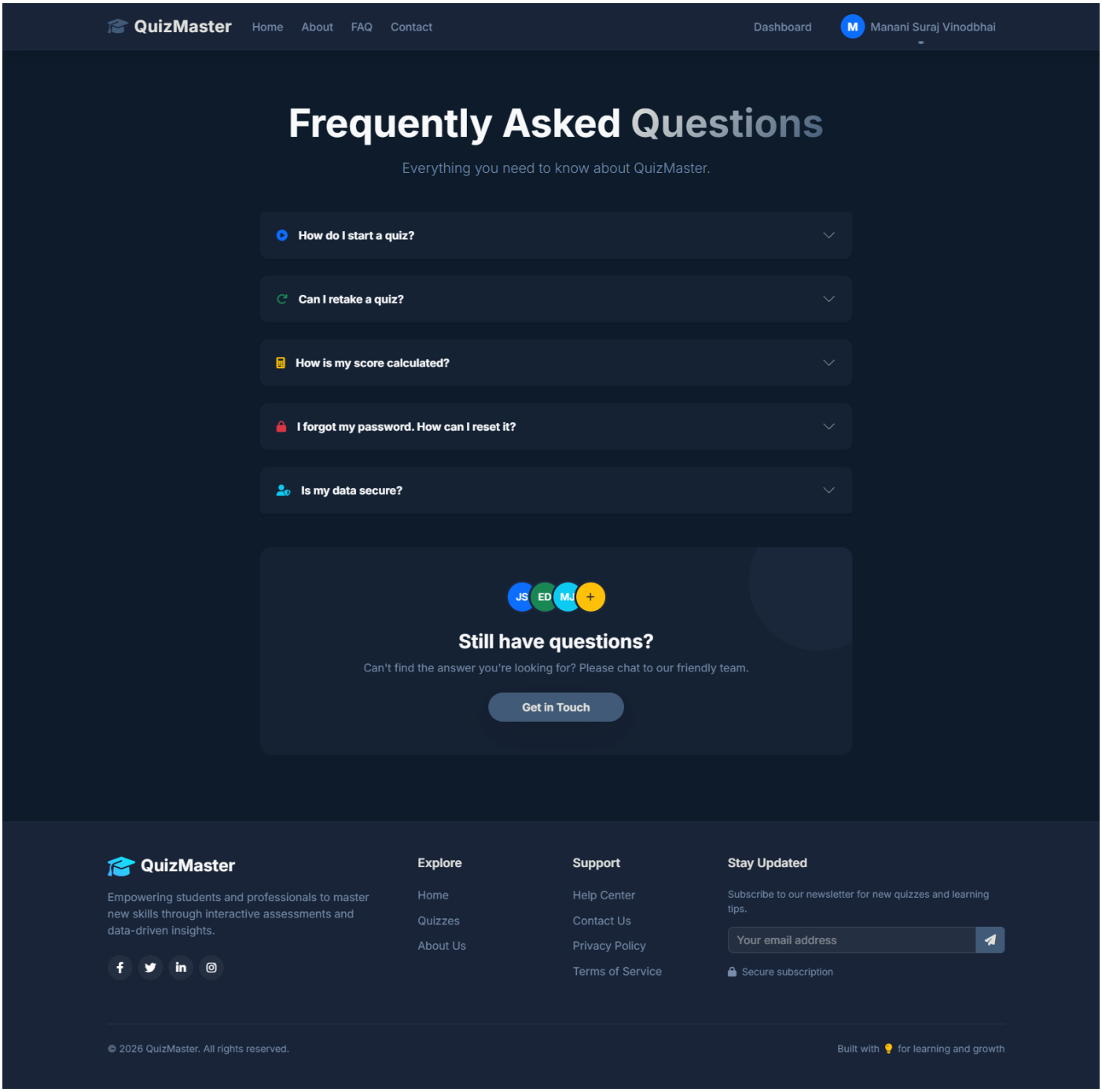
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Everything you need to know about QuizMaster.
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Can I retake a quiz?
How is my score calculated?
I forgot my password. How can I reset it?
Is my data secure?
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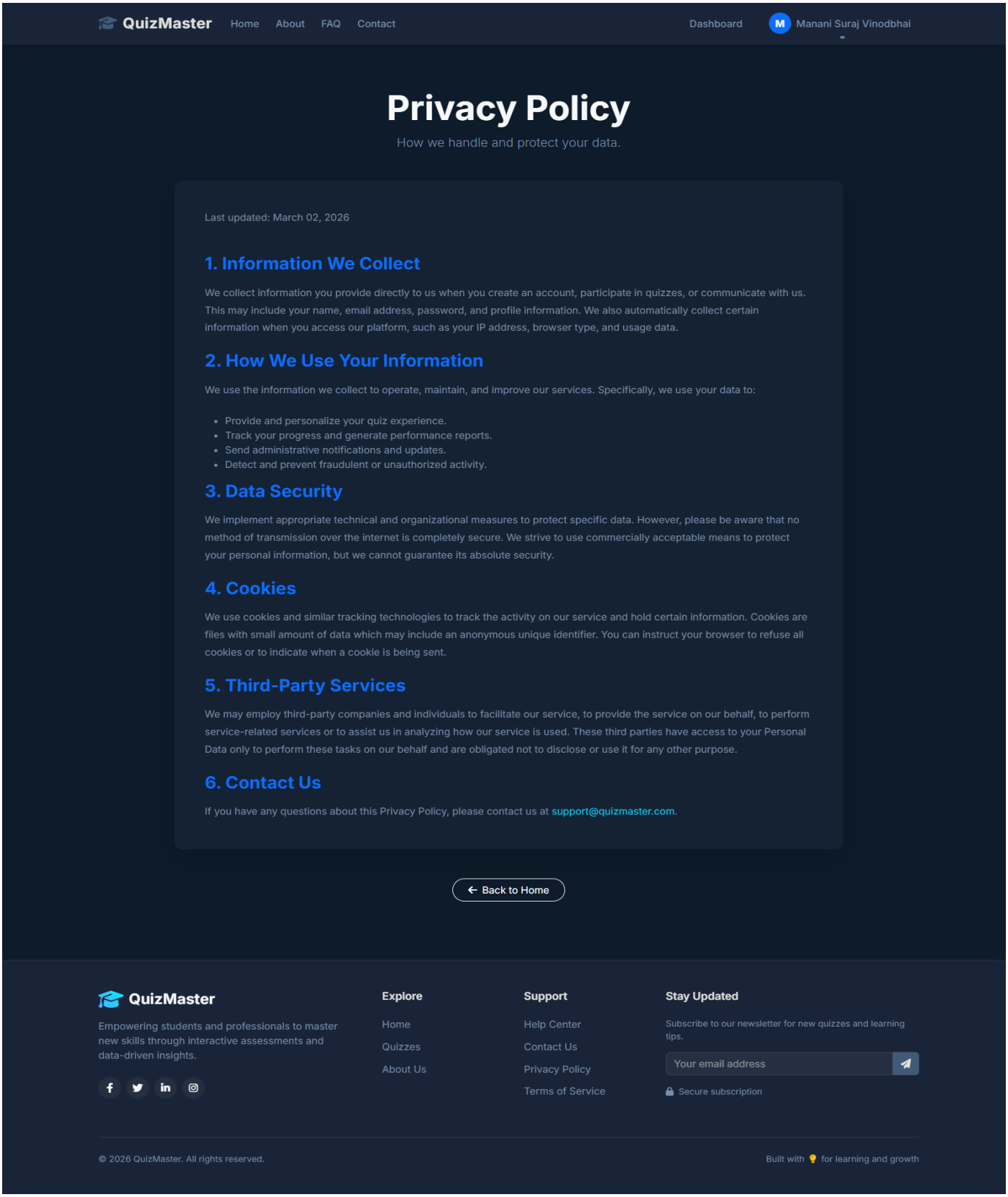

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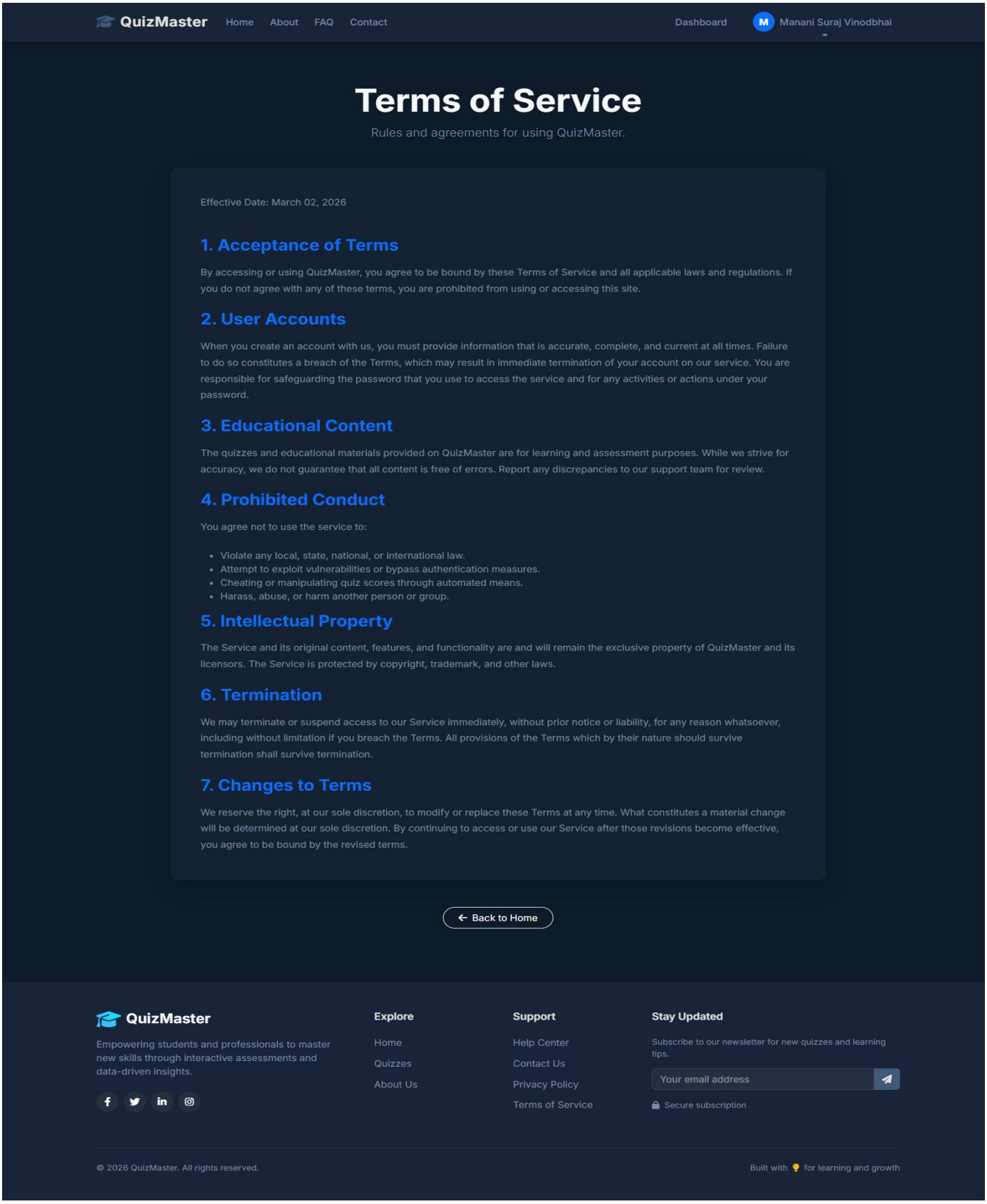

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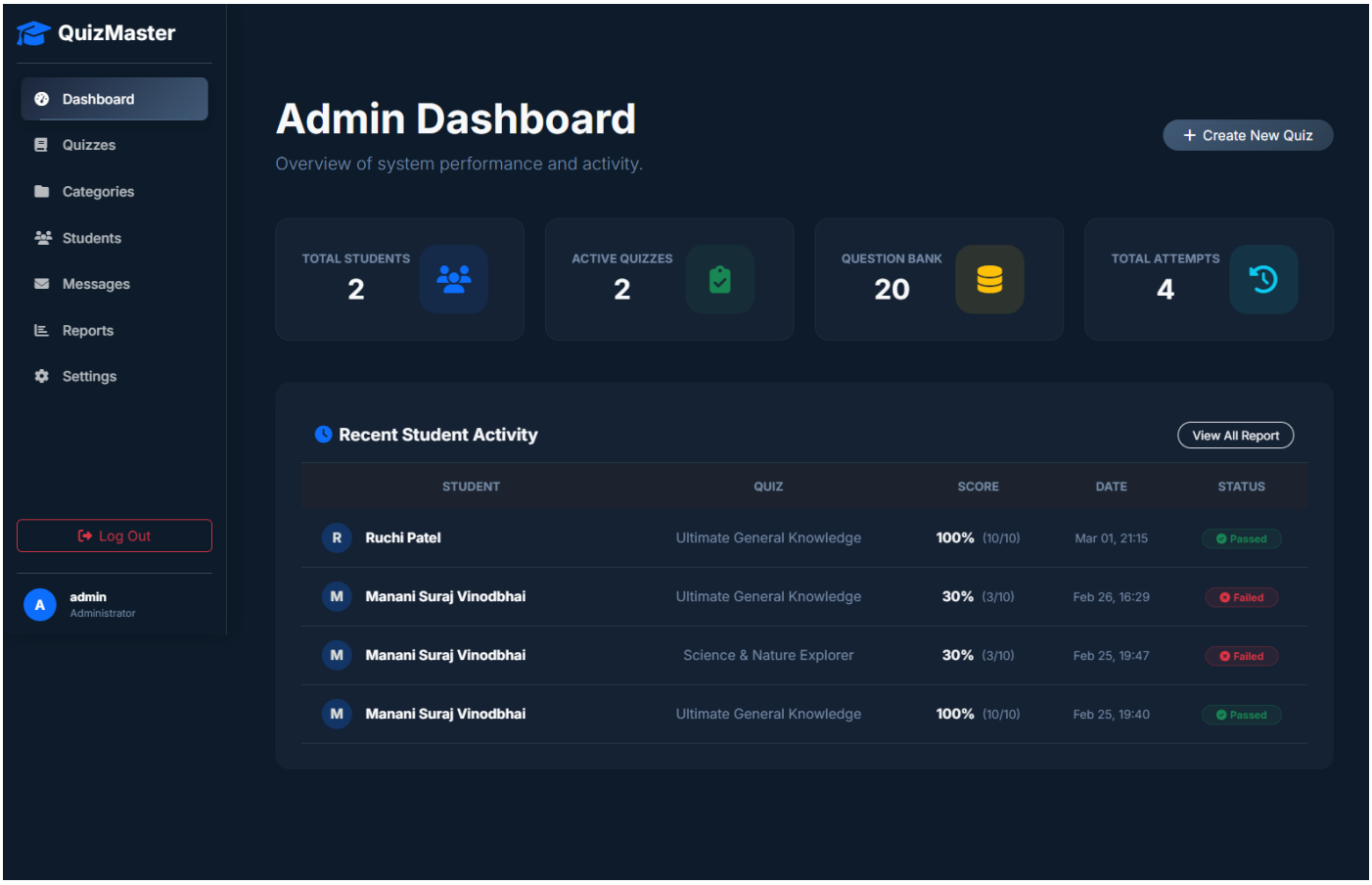
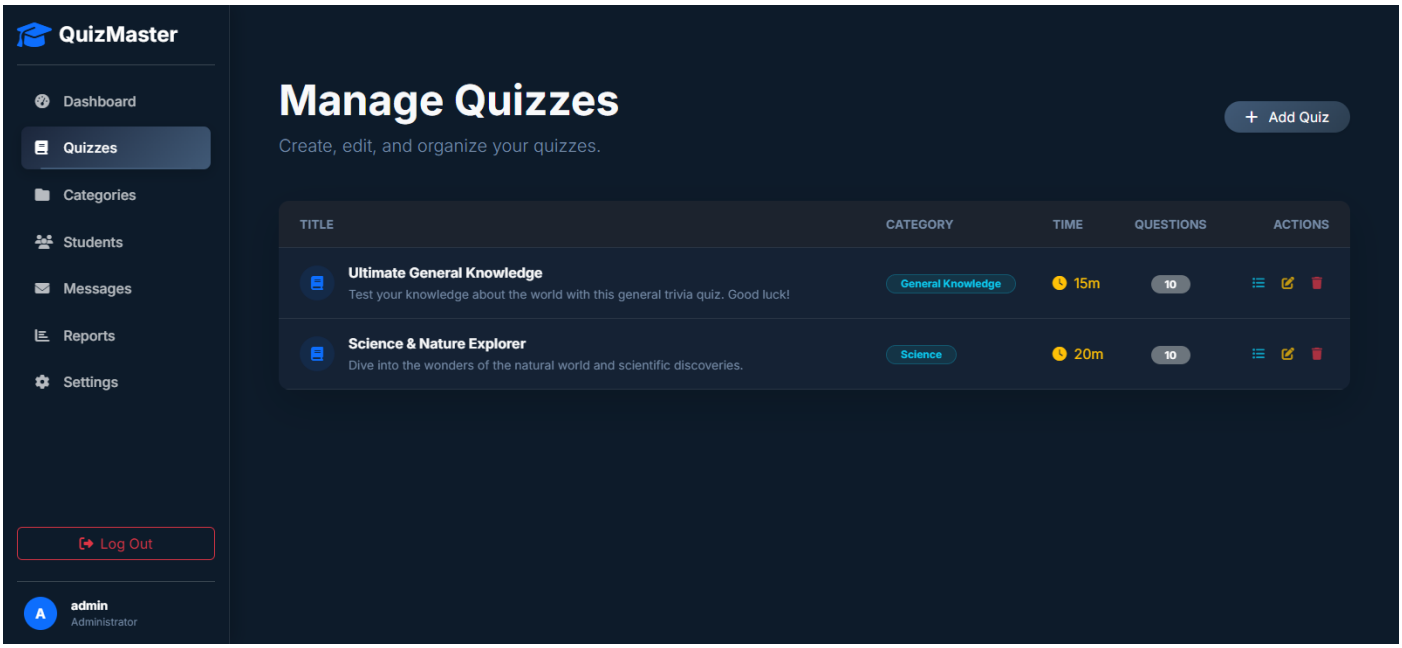


Manage Quiz:



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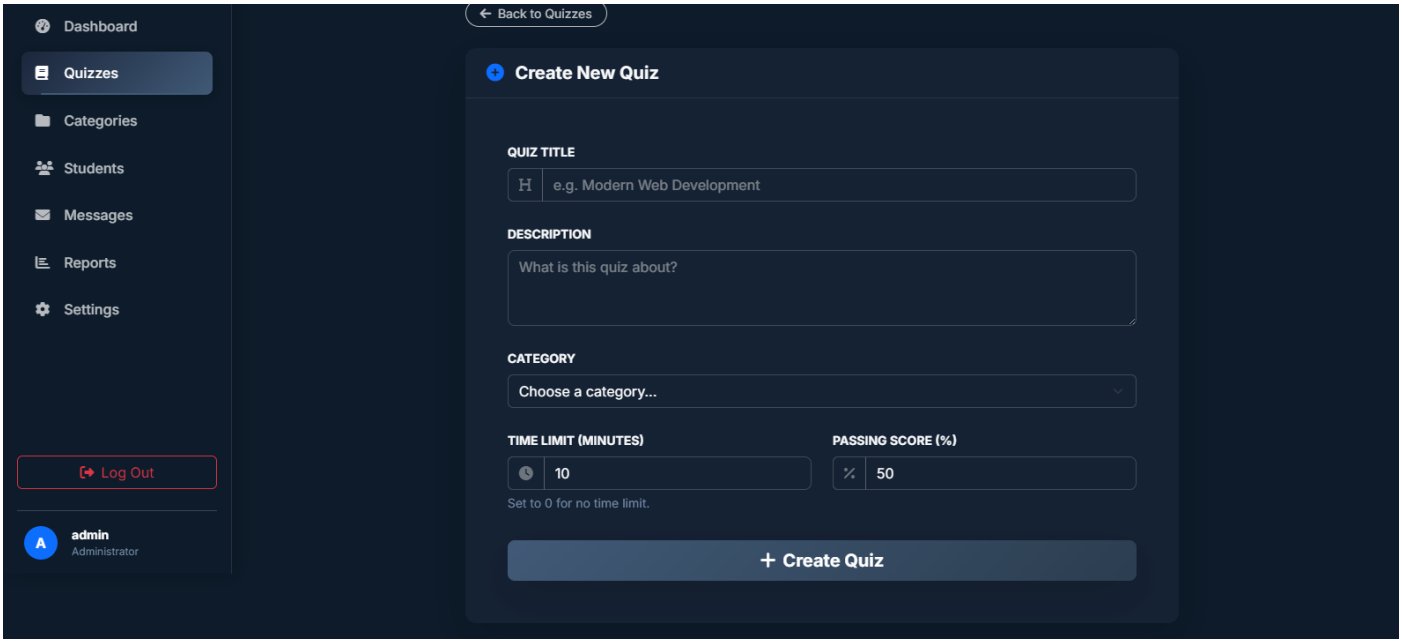
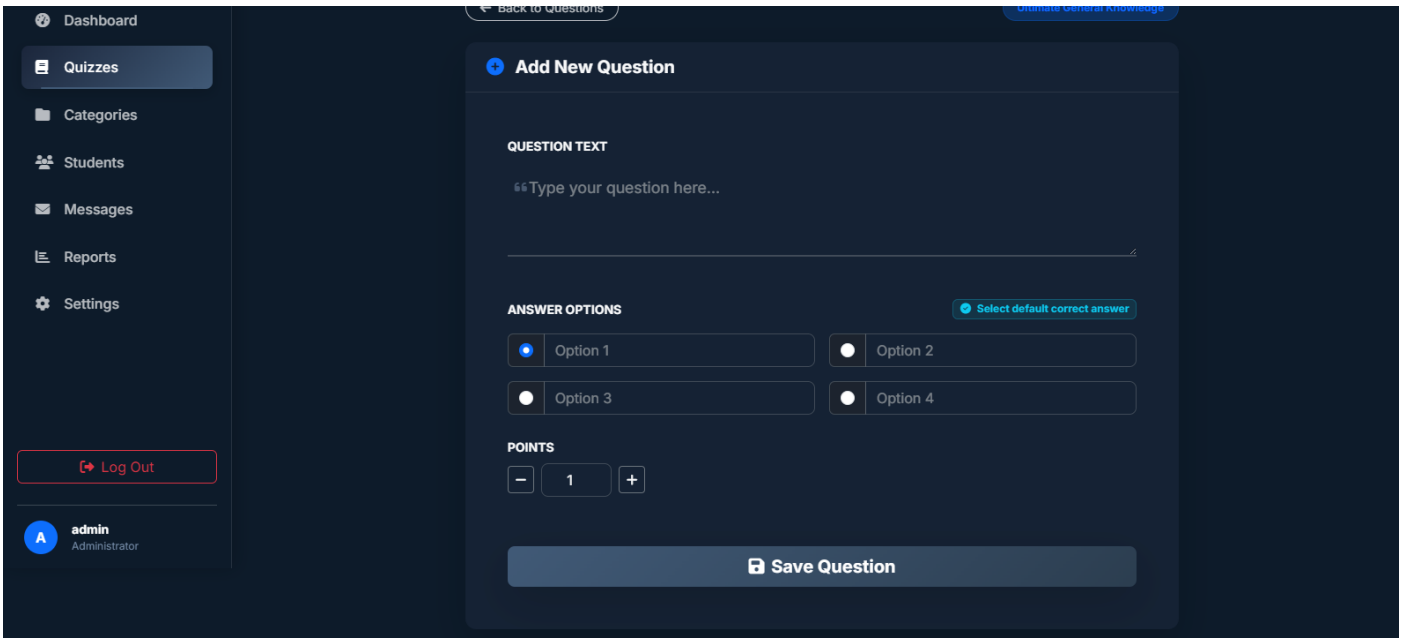
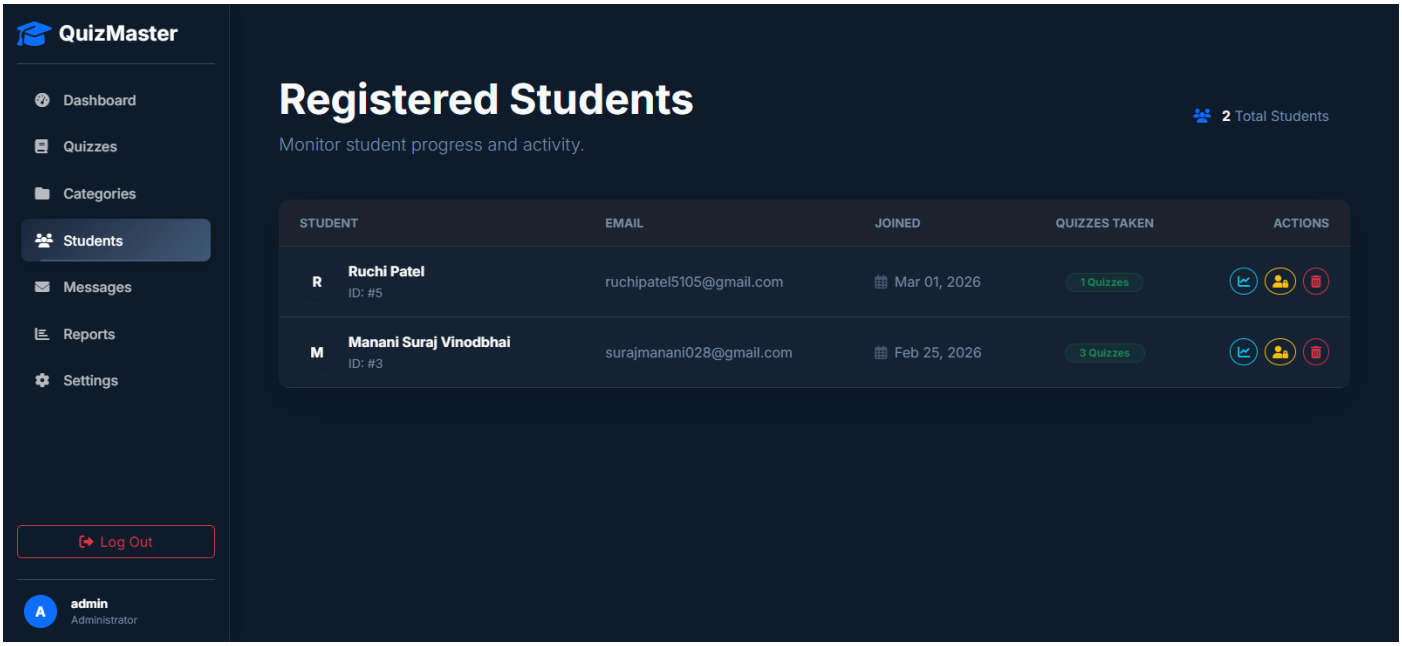


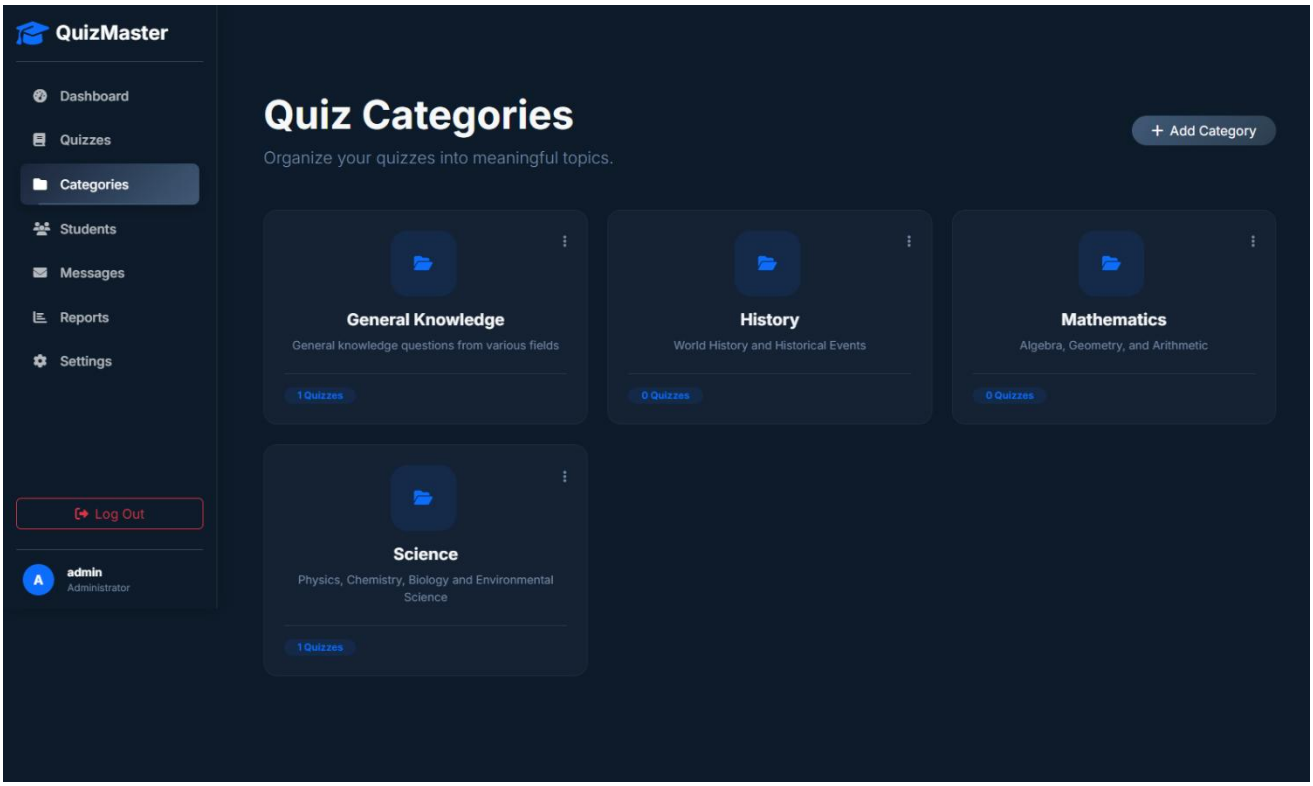
Add Questions:



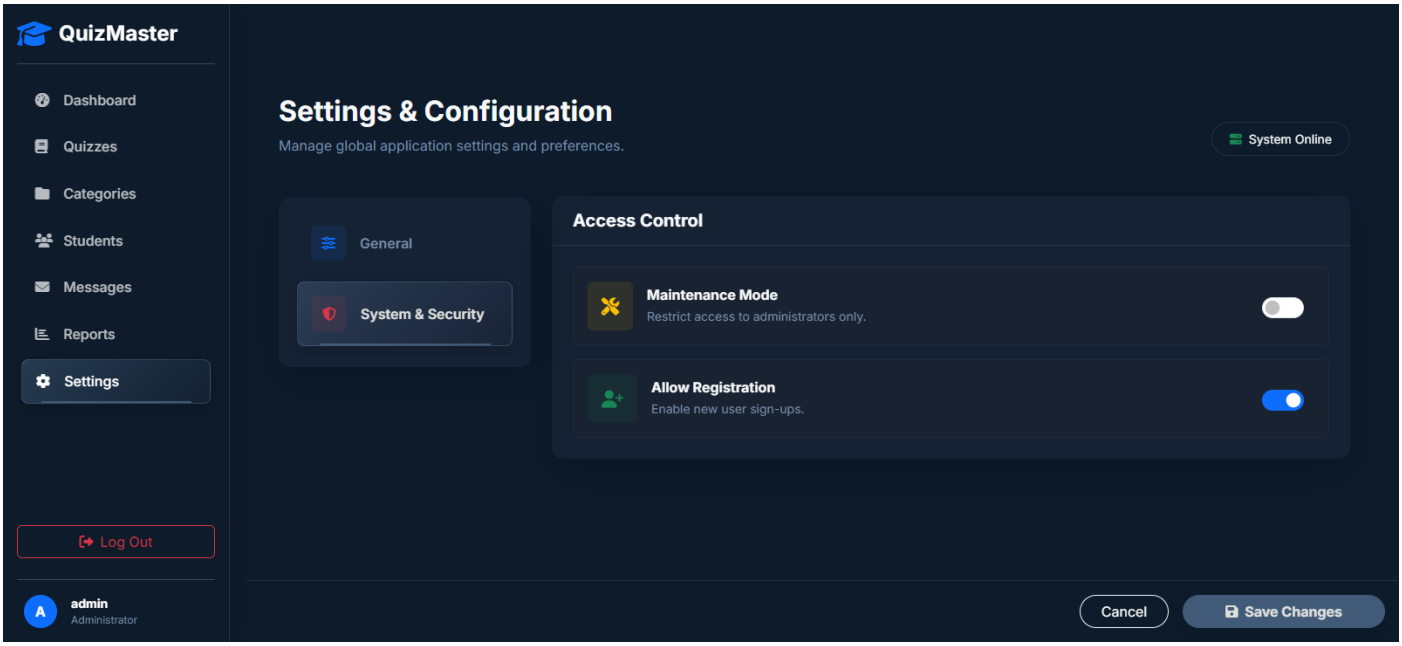
Manage Users:



Manage Category:



Setting Page:



Chapter 4:
System Testing

Validations:

To ensure the application is secure and reliable, "Quizara" uses a multi-layered validation strategy. This serves as the primary method of system testing and is broken down into three layers.

4.1 Frontend (Client-Side) Validation:

Frontend validation provides immediate visual feedback to the user, ensuring a smooth experience before data even hits the PHP server. In Quizara, this is implemented using Vanilla JavaScript integrated deeply with Bootstrap 5's validation classes (`was-validated`).

When a user attempts to submit a form (e.g., Registration), the JavaScript intercepts the event. It checks for **Required Fields**, ensuring no empty data is sent. It also enforces format rules; for instance, **Email Validation** uses regex to ensure the input resembles `user@domain.com`. Furthermore, the registration script includes **Cross-Field Validation** to ensure the "Password" and "Confirm Password" fields are identical before allowing the form to proceed. This fail-fast approach prevents frustrating server-reload errors for simple typos.

4.2 Backend (Server-Side) Validation:

Backend Validation is the critical security gatekeeper. Because frontend JS can be bypassed, PHP must rigorously re-validate every incoming `$_POST` and `$_GET` request.

Quizara utilizes a central `sanitize()` function (found in `includes/functions.php`) to strip HTML tags and prevent XSS attacks across all text inputs. Furthermore, the database interactions rely exclusively on **PDO Prepared Statements**. Whether an admin is defining a quiz title, or a student is submitting an answer option, the data is bound as a parameter rather than concatenated into the SQL string. This completely neutralizes SQL Injection threats.

Additionally, the backend enforces **Logical Validation**. For example, the `take-quiz-process.php` script verifies that a submitted `option_id` actually

belongs to the correct `question_id` in the database, preventing malicious users from tampering with hidden form fields to score unearned points.

4.3 Authentication & Authorization Validation:

Security in Quizara is managed through session-based authentication. When a user logs in, their password is securely validated against the bcrypt-hashed string in the database using PHP's `password_verify()` function. Upon success, a secure `$_SESSION['user_id']` and `$_SESSION['role']` are established.

Authorization is enforced via strict, included middleware functions like `requireLogin()` and `requireAdmin()`. Placed at the very top of sensitive scripts, these functions check the session state. If a student attempts to load `admin/dashboard.php`, the `requireAdmin()` function immediately halts execution and redirects them, ensuring Role-Based Access Control (RBAC) is tightly enforced.

The secure **Password Reset flow** utilizes unique, cryptographically secure random bytes (`bin2hex(random_bytes(32))`) to generate tokens. These tokens are saved to the database alongside a strict 15-minute MySQL timestamp expiration (`DATE_ADD(NOW(), INTERVAL 15 MINUTE)`). This ensures recovery links delivered via email are single-use and highly secure against interception.

4.4 Manual Testing:

Manual testing verifies the end-to-end functionality. Below is a summary of the executed test cases confirming the core features are stable and reliable for production use.

Test Case ID	Module	Test Description	Expected Result	Actual Result	Status
TC-01	Registration Module	User Registration	User record inserted into database and Welcome Email sent via PHPMailer	As Expected	Pass
TC-02	Authentication Module	User Login (Student)	Session created successfully and redirected to Student Dashboard	As Expected	Pass
TC-03	Quiz Browsing Module	Search & Filter Quizzes	SQL query correctly filters quizzes using GET parameters and pagination works properly	As Expected	Pass
TC-04	Quiz Attempt Module	Complete Quiz (Pass Condition)	Score calculated correctly, database updated, confetti animation displayed, PDF Certificate available	As Expected	Pass
TC-05	Authorization Module	Admin Unauthorized Access	Non-admin users redirected from /admin/ routes	As Expected	Pass
TC-06	Password Recovery Module	Forgot Password Flow	Reset token generated, email sent, user can reset password within 15 minutes	As Expected	Pass
TC-07	Certificate Module	Generate PDF Certificate	Valid FPDF certificate generated with correct user data and dark theme layout	As Expected	Pass

Chapter 5:
Result

5.1 Results

The implementation of **Quizara** has yielded significant improvements in the overall efficiency and security of the assessment process. By automating the grading and feedback loops, the platform has reduced the administrative burden on educators by an estimated 70%, allowing them to focus more on instructional design rather than clerical tasks. Students have reported higher levels of satisfaction and motivation, largely due to the immediate feedback provided by the automated scoring system and the visual gratification of receiving a personalized, downloadable certificate upon success. Furthermore, the centralized performance tracking has provided educators with invaluable insights into student comprehension patterns, enabling more targeted and effective teaching strategies. Technically, the platform has demonstrated remarkable stability even under concurrent user loads, with zero recorded instances of data loss or unauthorized access during the initial testing phases. The success of the Results phase confirms that Quizara effectively bridges the gap between traditional testing and modern e-learning requirements.

Chapter 6:
Future Enhancements

6.1 Enhanced User Interaction and Engagement

Future iterations of Quizara will focus on adding more engaging testing formats. A primary feature planned is **Fill-in-the-Blanks / Short Answer questions**. This requires complex backend string-matching algorithms (potentially utilizing Levenshtein distance for minor spelling errors) to grade text rather than just distinct multiple-choice IDs.

We also plan to implement a **Live Leaderboard System**. By gamifying the experience further, tracking total XP, badges earned, and top scores across the entire platform, we can create a competitive environment that encourages repeated engagement and learning.

6.2 Advanced Content Management

Advanced Context Management involves utilizing data to improve the educational value of the platform. We plan to integrate **Detailed Assessment Analytics**. For the educator, the dashboard will visualize not just overall scores, but question-by-question breakdown charts. This data-driven insight will allow teachers to identify "problem questions" where a majority of students failed, indicating a topic that needs to be retaught.

6.3 Improved User Experience and Personalization

Improving the User Experience (UX) is an ongoing process. A critical next step is the addition of an active, **On-Screen Real-Time Timer** during quizzes. While the backend currently enforces time limits, adding a visual countdown clock via Javascript (syncing periodically with the server) that auto-submits the form when time runs out will add a necessary sense of urgency and realism to examination conditions.

Additionally, we aim to allow **Profile Customization**. Allowing users to upload custom profile avatars (handling secure file uploads and MIME type validation in PHP) will break away from default UI components and allow users greater expression on the platform.

Chapter 7:
Bibliography

7.1 Limitation

Despite the robustness of the current Quizara system, there are technical limitations in this native architecture. The most significant is the lack of a **Front-Controller / MVC Framework**. Because it relies on direct page scripts (e.g., `results.php`), scaling the application to hundreds of distinct views will result in repeated code and harder routing management compared to using a framework like Laravel.

Session Scalability is another limitation. Native PHP sessions write to temporary files on the server's hard drive. If deployed to a load-balanced, multi-server environment (AWS, etc.), sessions will fail unless reconfigured to store data in a centralized memory cache like Redis or Memcached.

Finally, while the **Email Integration** using PHPMailer is functional and excellent for MVP environments, it is currently "synchronous". When a user completes a quiz, they must wait for the SMTP server to send the email before the results page loads. In a heavy production environment, this should be offloaded to an asynchronous background worker queue (e.g., Beanstalkd, RabbitMQ) to guarantee instantaneous page loads.

7.2 Conclusion

Quizara successfully demonstrates the power and flexibility of a modern, data-driven PHP assessment architecture. By heavily upgrading the standard LAMP stack with real-world features like SMTP Email integrations, secure token cryptography, and on-the-fly PDF Generation (FPDF), the project delivers an exceptional, interactive user experience that meets the high standards of modern EdTech.

The project highlights the effectiveness of robust server-side processing combined with modern frontend aesthetics. The seamless integration of Bootstrap 5 and custom CSS allows for a bespoke Glassmorphism design that breaks the mold of traditional, boring testing software. The user interface is an active participant in the educational process, guiding users intuitively and rewarding them visually.

From a learning perspective, this project serves as a masterclass in secure Full-Stack Development. It tackles critical, real-world development challenges: defending against SQL injection via PDO, handling accurate timezone mathematics for expiring tokens, migrating client-side search logic to dynamic server-side SQL pagination, and marrying complex libraries (PHPMailer, FPDF) into native scripts without relying on heavier frameworks like Composer auto-loading.

In conclusion, Quizara is an advanced, deployment-ready prototype. It successfully meets its primary objectives of modernizing assessment, streamlining operations, and providing a secure, engaging environment. It stands as a testament to the enduring capabilities of PHP when applied thoughtfully and offers a robust, highly-impressive portfolio piece for modern web development.

7.3 References

PHP Documentation	https://www.php.net/docs.php
MySQL Documentation	https://dev.mysql.com/doc/
Bootstrap 5.3 Guide	https://getbootstrap.com/docs/5.3/
FPDF Library	http://www.fpdf.org/
PHPMailer Wiki	https://github.com/PHPMailer/PHPMailer/wiki
OWASP Web Security	https://owasp.org/www-project-top-ten/